

State Power Attracts Talent: Evidence from China's Civil Servant Fever*

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Abstract

Why are government jobs so attractive to talented individuals in many non-democratic countries? We identify state power expansion as a key factor. We analyze the impact of China's recent value-added tax reform, an expansion of taxation power, on citizens' preferences for tax administration positions, as revealed by participation in the National Civil Service Exam. Using a difference-in-differences approach that leverages pre-reform subnational variation in tax composition and a dataset of 166,012 government job openings from 2010 to 2021, we find that the reform attracted more and higher-ability individuals to tax-related state positions. The effect is particularly strong for positions involving greater regulatory power or those located in prefectures with higher graft opportunities. An original survey reveals that exam takers perceive increased power and benefits in tax agencies after the reform. Evidence from Chinese General Social Surveys suggests the talent drawn to the state likely comes from the private sector.

Keywords: Non-democracy, state power, civil service exam, tax reform, VAT, talent recruitment

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1 Introduction

In *Democracy in America*, Alexis de Tocqueville observed, “it is a constant fact that in our day, in the United States, the most remarkable men are rarely called to public offices” (Tocqueville, 1835/2000). Tocqueville believed that in democracies, civil servant jobs are not particularly enticing to society’s brightest minds. This is because democratic institutions channel the public’s aversion to elitism, particularly in government offices. The instability of government careers due to elections and term limits further dissuades accomplished individuals from pursuing them. In stark contrast, citizens in less democratic or non-democratic countries often exhibit a strong preference for government jobs. In Bangladesh, such jobs are considered privileges and significant opportunities. The introduction of a quota system limiting students’ access to civil servant jobs in 2024 sparked nationwide protests and violent crackdowns, resulting in 200 deaths and thousands of injuries and arrests.¹ Similarly, in China, millions of individuals take the Civil Service Exam annually in hopes of securing a government position. For instance, in 2021, the most sought-after jobs attracted a staggering 20,602 applicants competing for a single position.² At top Chinese universities such as Tsinghua and Fudan, 50%–70% of graduates land positions within the state sector every year.³ China and Bangladesh are not isolated cases. Similar preferences for government jobs are evident in Singapore, Iran, Vietnam, Russia, Egypt, and many other non-democratic countries. What explains non-democratic governments’ ability to attract talented individuals to the state?

In this paper, we identify state power as a key factor attracting talent to the state apparatus in non-democratic regimes. Following Way (2005), we conceptualize state power along three dimensions: control, scope, and size. Control refers to top officials’ ability to enforce obedience from subordinates to suppress opposition and maintain stability, enabled by centralized fiscal and administrative power. Scope encompasses the range of issues under state influence, with such regimes expanding control over the economy and society to stifle opposition and limit private sector autonomy. Size involves the nationalization of key sectors to expand state reach and secure internal resource control. Together, non-democratic regimes often expand state power by enhancing central

¹Behind Bangladesh Protests, Rage Over Inequality. *New York Times*. <https://nyti.ms/4lGRpcf>

²The Most Competitive Position Has a Ratio of 20,602:1, *Soho.com*. <https://bit.ly/4iqTw9H>.

³For more details about Tsinghua and Fudan graduates, see Figure A1. Li, Meng, Xiong, and Cook (2023) also show that a majority of Chinese college students desired employment in the state sector in recent years. Bai, Jia, Li, and Wang (2025) further show that China’s most talented individuals gravitate toward the state sector instead of launching their own firms.

fiscal capacity and broadening the scope and size of control for political survival. Although the broader literature often discusses these processes under the heading of *state building*, our argument focuses more specifically on *state power expansion*, by which we mean the expansion of the state's fiscal, coercive, and regulatory reach in non-democratic regimes.

Unlike state building in liberal democracies, which often emphasizes representative institutions, the rule of law, and accountability mechanisms that limit the opportunities for public officials to benefit from state power, state power expansion in non-democratic regimes invariably yields tangible rewards for those within the system. For example, enhanced central fiscal capacity enables the regime to allocate more resources, offering state employees better material benefits. The expansion of state control over the economy and society, often at the expense of the private sector, not only makes careers in the state sector more secure and promising but also grants state employees opportunities to secure privileges and extract graft from society, especially given that checks on the authority of government officials are limited in these regimes (Frye & Shleifer, 1997). Perceiving the benefits of government jobs in a powerful state, talented citizens are thus attracted to the state apparatus. This logic may also extend to weak or illiberal democracies, where limited institutional constraints allow state power expansion to yield comparable private returns for public officials.

The ability of a powerful state to attract talent is especially strong when combined with merit-based recruitment. Leaders in non-democracies have long utilized the state sector to co-opt societal actors (Ang, 2016; de Mesquita, Smith, Siverson, & Morrow, 2005; H. Liu, 2023; Magaloni, 2006; Rosenfeld, 2017; Svolik, 2012). However, if the hiring of state employees relies on connections, clientelism, or nepotism (Calvo & Murillo, 2004; Fukuyama, 2014; Hassan, Larreguy, & Russell, 2023; Jiang, 2018), a powerful state may draw more individuals to the state sector but not necessarily those of high ability. Faction-based political selection and the tendency to prioritize loyalty over competence can also impede these regimes' capacity to attract high-ability talent from society (V. Shih, Adolph, & Liu, 2012; Zakharov, 2016). Nevertheless, if a regime can implement merit-based recruitment, even partially, in its political selections—as has been observed in many countries such as China (Landry, Lü, & Duan, 2018; H. Liu, 2023) and Ghana (Brierley, 2021)—then a powerful state is likely to attract a significant amount of talent to the regime.⁴

⁴Merit-based recruitment through standardized exams can coexist with systematic corruption: talented individuals may use their skills corruptly when the system permits it. As the literature shows, checks and balances, rather than merit-based recruitment alone, are what effectively curtail corruption (Treisman, 2007).

To provide empirical evidence, we examine the impacts of China’s value-added tax (VAT) reform—a recent expansion of state taxation power—on citizens’ preferences for tax administration positions revealed through participation in the National Civil Service Exam (NCSE). Between 2012 and 2016, China implemented the most significant tax reform in the past two decades by effectively replacing the business tax (BT) with the VAT. The stated purpose of the VAT reform was to ease the tax burden on service industries, but it also significantly increased central tax bureaus’ tax-extraction capacity (C. Zhang, 2021, p. 53). This enhanced central fiscal capacity is a critical feature of state power expansion. Moreover, VAT has a self-enforcing nature because it incentivizes downstream client firms to request invoices from suppliers to reclaim the VAT they have paid. This chain of tax enforcement makes tax evasion more challenging for private enterprises (Pomeranz, 2015), especially when compared to the BT system, where high nominal tax rates in China made evasion extremely prevalent (C. Zhang, 2021, p. 77). By enhancing tax enforcement through the VAT system over a much larger number of enterprises that previously paid the BT, the VAT reform also expanded the scope of state control—another critical feature of state power expansion.

China’s NCSE has been a merit-based system for recruiting government officials through standardized exams open to all Chinese citizens since 1994 (H. Liu, 2023). In 2003, the number of applicants doubled from the 2002 figure to 125,000 and has kept rising ever since. In 2021, over 2.12 million participated in the exam, each with only a one-in-68 chance of securing a government job. The sheer number of applicants and the exam’s intense competition underscore the growing allure of government jobs in recent years. This phenomenon is known in China as the “civil servant fever”, or “gong wu yuan re (公务员热).” Because different positions attract varying numbers of exam takers, the level of competition for each position reflects citizens’ preferences for specific government jobs. We expect the VAT reform to increase the attractiveness of tax administration positions, as the reform signals an expansion of state taxation power.

We use a difference-in-differences (DiD) strategy to causally identify the effect of the VAT reform on the popularity of tax administration jobs and applicants’ ability level. We exploit variations in pre-reform BT to VAT ratios across 289 Chinese prefectures and sort them into a high BT-VAT group and a low BT-VAT group based on the pre-reform mean BT-VAT ratio. Our DiD strategy compares these two groups before and after 2016, the year when the VAT reform was fully

rolled out. The rationale is that the VAT reform would have increased the power of tax agencies in prefectures with high BT-VAT ratios more than that of those with lower ratios.

We collect a novel dataset that combines all NCSE job postings from 2011 to 2021, a total of 166,012 job posts. The key outcome measures are the number of applicants and the applicant-to-recruitment (App-to-Recruit) ratio, which reflect the popularity and competitiveness of a position, respectively. We also use the interview cutoff score to measure the ability level of applicants, as a higher cutoff score indicates a better candidate pool. We divide the data into two subsamples: 1) job openings affiliated with the State Administration of Taxation (the SAT sample), and 2) job openings outside the State Administration of Taxation (the Non-SAT sample). We primarily focus on the SAT sample, while the non-SAT sample serves as a placebo test and enables us to examine the spillover effects of power expansion in tax administration.

We find that, among the SAT sample, the VAT reform increased the number of applicants, the App-to-Recruit ratio, and the interview cutoff score by 20.7%, 18.4%, and 1.1% of their respective means in the high BT-VAT group relative to the low BT-VAT group. Since a one-point change in test scores can eliminate a substantial number of candidates, the ability level of the finalists increased significantly. These effects remain robust when using a continuous treatment measure of BT-VAT ratio, controlling for time-varying prefecture-level covariates, and applying different methods to assign high-low BT-VAT groups. Conversely, no significant effects of the VAT reform were observed among the Non-SAT sample, suggesting that the talent attracted to SAT positions is unlikely to come from the pool of applicants for other government jobs. As a central agency, the SAT may draw talent from the pool of applicants for Local Administrations of Taxation (LATs) if the reform shifted talent allocation through power centralization. To address this concern, we first use data on the Zhejiang Provincial Civil Service Exams and show that the VAT reform also increased applications to the LATs.⁵ We further analyze self-reported employment data from six waves of the Chinese General Social Survey and show that the VAT reform reduces the likelihood of an individual working in the private sector compared to the public sector, suggesting that the reform drew talent from the private sector into the broader state sector, beyond tax agencies. These results together suggest that state power expansion through the VAT reform likely attracts talent to

⁵Provincial Civil Service Exam data are limited because provinces differ in the level of information disclosure, leaving many years and regions without records. However, we were able to collect four years of data for Zhejiang Province to supplement our main analysis of the NCSE.

the SAT from broader society rather than from other state agencies.

State building helps attract talent to the state sector in the short run because citizens can easily *perceive* the increased power and benefits associated with government jobs following the expansion of state power. However, the long-term impact of state building on talent allocation may be ambiguous and contingent on specific contexts. If a non-democratic regime expands state power at society's cost, this can enhance the allure of government jobs even in the long run. On the other hand, state building may facilitate economic growth in countries where the state was previously too weak to offer basic public goods, infrastructure, and social order. If, in the long run, state building proves more beneficial to the private sector than to the public sector, it could redirect talent away from the state sector. Our evidence appears to support the former scenario in China, as five years following the VAT reform, there is a sustained influx of talent into state tax agencies.

One key mechanism by which state power expansion attracts talent is the lack of constraints on state power and associated graft opportunities. To explore this mechanism, we first categorize jobs into those with greater power, such as tax inspection jobs, and those with less power, such as office and front desk positions. We find that the VAT reform led to significantly larger increases in the attractiveness of jobs with more power. We further construct a prefectural-level measure of corruption opportunities using the number of investigated officials prior to the 2012 anti-corruption campaign. We find that the VAT reform attracts more and better talent in prefectures with greater corruption opportunities.

We further conducted an online survey of 3,700 citizens and identified 788 eligible respondents who have either taken or plan to take the NCSE. We find that most individuals aiming for SAT jobs are aware of the VAT reform, and the vast majority perceive an increase in the power and benefits of SAT jobs following the reform. Regression analyses find that respondents with higher social status and greater educational attainment are more likely to associate the reform with increased power for SAT agencies. These findings align with our argument that state power expansion attracts societal elites into the state in pursuit of power and privileges—contrasting with De Tocqueville's observations in America.

2 State Power Expansion and Talent Attraction

State building is one of the most widely discussed but contested concepts in political science (Centeno, Kohli, & Yashar, 2017). The literature often uses state capacity interchangeably with power, strength, and institutions, with definitions emphasizing implementation, policy scope, coercive ability, or bureaucratic competence (e.g., Dahl, 1957; Evans & Rauch, 1999; Fukuyama, 2004; Lowi, 1964; Mann, 1984; Migdal, 1988; Weber, 1919). In this paper, we define state building as the centralization of coercive power and resource extraction, which can then be used to maintain political order and/or provide public goods (Centeno et al., 2017).

The literature on state building largely emphasizes the European experience characterized by the development of representative institutions, the rule of law, constraints on executive power, and a professional, merit-based bureaucracy (Besley & Persson, 2009; Fearon & Laitin, 2003; Herbst, 2000; Jackson & Rosberg, 1982; Migdal, 1988; Slater, 2010; Tilly, 1985; Weber, 1919). Scholars attribute the association between state building and democratic institutions to factors such as political fragmentation following the fall of the Roman Empire (Stasavage, 2020), the weak bargaining power of rulers relative to elites (North & Weingast, 1989), the influence of Christianity (Fukuyama, 2014, pp. 229–261), and sovereign borrowing practices (Cox, 2016; Saiegh, 2005). Even theories emphasizing interstate conflict and resource extraction note that European states' selective extension of protection to different social classes “constrained the rulers themselves, making them vulnerable to courts, to assemblies, to withdrawals of credit, services, and expertise” (Tilly, 1985, p. 186).

Democratic state building does not necessarily make government jobs attractive to society's most talented individuals. As Tocqueville noted, the equalization of conditions in democratic societies could lead to “tyranny of the majority” and mediocrity in governance (Tocqueville, 1835/2000). Specifically, citizens' disdain for elites and elitism could be channeled by electoral systems and potentially affect the attractiveness of government positions for highly talented individuals. Moreover, factors such as the electoral process, transparency, and checks and balances often limit officials' opportunities to benefit from public offices (Eggers & Hainmueller, 2013).⁶ This aligns with the concept of the Weberian bureaucratic state, which emphasizes a professional, neutral,

⁶Private gains from public office are possible in democracies, but unethical behavior among politicians and bureaucrats is typically more restricted in mature democracies than in non-democracies.

and competent bureaucracy governed by codified rules (Evans & Rauch, 1999; Weber, 1919). In other words, democratic state building can result in diminished incentives for talented individuals to pursue careers in the public sector. This view is consistent with recent empirical evidence on a relatively lower degree of meritocracy among U.S. federal employees compared with Chinese civil servants (Boittin, Distelhorst, & Fukuyama, 2016) and public skepticism of science in democracies (Jiang & Wan, 2023).

While much of the existing literature focuses on democratic state building, strong states also emerge within non-democratic regimes without leading to democratic development (Y. Huang, 2023; Y. Wang, 2022a, 2022b). Examples of powerful states can be found in the Soviet Union/Russia, China, Saudi Arabia, the United Arab Emirates, Singapore, and others. In these contexts, state building is aimed at increasing the power and resources of incumbent governments to ensure their survival (Way, 2005). As Way argues, state power in non-democracies often expands through three dimensions: control, scope, and size. Control refers to the degree to which top officials can depend on subordinates to suppress opposition and manipulate elections. This is influenced by fiscal health and patronage, both of which typically require the centralization of fiscal and administrative power. Scope pertains to the range of issues leaders exert control over. Greater state control over the economy can stifle opposition, while a strong private sector often supports independent media and opposition movements. To expand their scope, non-democratic regimes often implement policies that limit private sector autonomy and increase state intervention in economic and social spheres. Size impacts fiscal stability and the regime's exposure to Western democratization pressures, especially in the post-Cold War period. This often involves nationalizing economic sectors to expand state reach and secure internal resource control. In short, the core logic of state building in such regimes is to expand state power for political survival.

These key features of state power expansion in non-democracies, particularly its expansion in scope, sharply contrast with democratic state building, which is characterized by limited government rather than pervasive control. As Fukuyama (2004) observes, the United States operates under a system of limited government that carefully restricts the scope of state activity. Within that restricted scope, however, the state wields substantial power—not merely on paper but in its capacity to enact and enforce laws and policies effectively. This effectiveness is achieved largely through citizens' voluntary compliance with laws due to the regime's legal-rational legitimacy and

adherence to the rule of law (Weber, 1919).

By contrast, non-democratic states seek to expand both the scope and size of the state to control society and the private sector, prioritizing regime stability and survival. Moreover, without the rule of law, citizens often lack voluntary compliance, prompting rulers to further expand the scope of state activities to compensate for the lack of effectiveness and to rely on coercive measures for enforcement. These differences imply that state power expansion in non-democratic regimes is not accompanied by limited government or executive constraints; instead, it often involves the concentration of resources to the state sector while systematically weakening the private sector and societal forces to consolidate state control. In this sense, state power expansion in such settings entails not only absolute state power expansion but also changes in relative state power—how strong the state is compared to the private sector. In a weak society with an underdeveloped private sector, a relatively stronger state, even if weak in absolute terms, can still maintain regime stability. But when the private sector grows rapidly, as in China during the reform era, the regime faces strong incentives to expand state power—both in scope and size—to contain the rising influence of the market. Consequently, state power expansion in non-democratic regimes often involves not just building capacity but also deliberately constraining private-sector development to maintain the state's relative dominance. This is a crucial reason why state power expansion draws talent to the state sector from society in non-democratic regimes.

State power expansion makes government jobs particularly attractive, in part because power in these regimes is often unconstrained. Rulers use the state sector to co-opt both the masses and elites (de Mesquita et al., 2005; Magaloni, 2006; Svolik, 2012), deter opposition and rebellion by keeping potential rivals close (Gandhi & Przeworski, 2007), and establish patron-client networks to secure loyalty and maintain power (Blaydes, 2010). A state with significant power and resources can create entry barriers and grant officials substantial authority, privileges, and opportunities for personal enrichment, often through corrupt and opaque practices (Haber, 2006). The lack of power constraints in these systems often leads to economies marked by corruption, informal practices, and high barriers to entry for those without political connections (Z. Chen, Jin, & Xu, 2021; Fisman, 2001; Manion, 2004; V. C. Shih, 2008; E. H. Wang, 2022). Scholars have long linked corruption to non-democratic rule (e.g., Ang, 2020; Gerring & Thacker, 2004; Treisman, 2007). But precisely for this reason, government jobs can be more lucrative than those in mature democracies, which

can further draw in talented individuals to the state sector.

For citizens in non-democratic regimes, an expansion of state power relative to the private sector provides government jobs with many advantages. First, a stronger state is more effective at extracting taxes and resources for the regime, allowing it to offer better material benefits to state employees. Second, the increased regulatory power affords state employees more privileges in society and potentially more opportunities to engage in graft. Third, the hierarchical structures and relative stability of these governments, characterized by the absence of regular power alternations, provide talented individuals with longer career paths and brighter career prospects. Fourth, when the power and regulatory capacity of the state grow, state employees enjoy higher social status, as the state sector possesses more discretionary power over society and the private economy. Finally, expanding state power also increases job security in the public sector as it signals that the government is too big to fail. While some of these factors might be applicable in democracies during phases of state power expansion, they are certainly more salient in non-democratic systems. Importantly, these constitute immediate benefits of government employment that citizens can easily perceive from observing state power expansion. Thus, state power expansion should increase talent attraction, at least in the short run.

H1: State power expansion increases the number of applicants to government jobs.

H2: State power expansion increases the competitiveness of applications for government jobs.

Of course, if the job recruitment system of the state sector is biased toward loyalty and patron-client connections rather than pure talent or ability, the expansion of state power would not attract higher-ability individuals to the state sector. However, if the expansion of state power is paired with a merit-based recruitment system, we would expect it to increase the ability level of applicants for government jobs. Here, ability refers to performance or cognitive skills, including test-taking capacity, and does not imply moral virtue. In other words, high-ability individuals may be intelligent yet also greedy, corrupt, or power-seeking.

H3: State power expansion combined with a merit-based recruitment system increases the ability level of applicants to government jobs.

An important mechanism through which state building in non-democratic countries makes government jobs more attractive to citizens is the often-unconstrained power and regulatory prowess

associated with those jobs. Since some jobs involve greater regulatory authority than others, state power expansion will have varying effects on jobs with different levels of power and regulatory prowess.

H4: State power expansion increases the attractiveness of government jobs offering greater regulatory power compared to those offering less power.

State power expansion can attract talent to the state sector in the short run by offering immediate and visible benefits of government employment, but its long-run effects are context-dependent. In some settings, stronger state capacity may promote private-sector development by securing property rights, improving infrastructure, and reducing crime (Dincecco & Katz, 2016; Fukuyama, 2014), making private-sector jobs more attractive over time. In others, a growing private sector may become politically threatening, prompting non-democratic regimes to reassert control through regulatory tightening and expanded state intervention. China illustrates this dynamic trajectory: state capacity initially enabled private-sector growth, but the regime later expanded state power to maintain its relative dominance. When state power increases relative to the private sector, sustained talent flows toward the state sector are more likely to occur. Thus, we propose the following research question (RQ):

RQ: What is the long-run effect of state power expansion on the popularity of government jobs in a particular country?

Two scope conditions are crucial for observing state power expansion and talent attraction in non-democratic regimes. First, the regime must have both the incentives and capabilities to expand the power of the state. Thus, our argument may not apply to rentier states where rulers can attract talent through resource rents without the need to expand state power. Similarly, it may not apply to failed or failing states where rulers lack the ability to increase state power or exert control over society. Second, the regime must implement at least partially merit-based recruitment for political positions. Without this, the expansion of state power is unlikely to attract high-ability individuals to the state sector, particularly in countries where cronyism, nepotism, and connections predominate in official recruitment. Of course, in weak democracies where constraints on opportunistic behavior by public officials are limited, state power expansion may have similar talent-attraction effects.

The theory suggests that unconstrained state power creates opportunities for graft that can attract talent to the state sector, making corruption deeply entrenched in such systems. Because rampant corruption undermines regime legitimacy, central authorities often launch anti-corruption campaigns to contain it, which can temporarily reduce the attractiveness of government jobs (Lai & Li, 2024). However, because these regimes depend on expanding state power for political survival and lack incentives to constrain that power through effective checks and balances, corruption tends to resume once anti-corruption efforts subside, along with the inflow of talent into the state sector. In this sense, the logic closely resembles what Manion (2004) describes as “corruption by design.”

3 Institutional Background

This section first explains why the VAT reform is a suitable case for this study. It then offers a brief overview of the expansion of state power in China over the past two decades, focusing particularly on the VAT reform and the phenomenon of “Civil Servant Fever.”

3.1 Value-Added Tax

In this paper, we focus on value-added tax (VAT) reforms as a form of state power expansion. A VAT is a broad-based consumption tax levied at each stage of production, with firms able to credit taxes paid on intermediate inputs. Because downstream firms must obtain invoices from upstream suppliers to claim input tax credits, VAT creates supply-chain transparency and mutual monitoring that constrains tax evasion. It is therefore attractive to governments seeking stable and enforceable revenue, particularly where income or sales taxation is difficult (Keen & Lockwood, 2010). VAT reforms therefore reflect a significant expansion of state power over the private sector, a feature that closely matches the concept of state power expansion discussed above.

Although all regimes may adopt VAT, often in response to fiscal pressures, economic crises, or the need to combat tax evasion (Keen & Lockwood, 2010; Mahon, 2004), their subsequent uses of VAT revenue differ markedly across regime types. In Western democracies, VAT revenues constitute a major source of funding for expansive welfare states (Kato, 2003; Wilensky, 2002). Democratic accountability facilitates redistribution through social transfers, while constitutional liberalism constrains public officials from privately benefiting from tax revenues. By contrast, non-democratic regimes typically lack strong mechanisms of accountability for redistribution, and

VAT revenues are more often used to strengthen the state apparatus, control society, and reward state employees. The absence of effective constraints on tax and regulatory officials allows expanded fiscal capacity to translate into greater material benefits for these officials, regulatory privileges, and opportunities for rent extraction. Thus, a VAT reform is more likely to increase the attractiveness of taxation jobs in these regimes.

3.2 State Power Expansion and the VAT Reform in China

In the early reform era, China deliberately reduced direct state control over the economy in order to promote growth. The state liberalized markets, privatized many state-owned enterprises (SOEs), opened space for private and foreign investment, and decentralized fiscal authority to subnational governments (e.g., [Qian, 2017](#)). For example, in 1991, SOEs held a 68% share of industrial output, but this figure fell to 35% in 2003 ([Naughton, 2018](#)). This process of marketization and deliberate reduction of state power allowed the rise of the private sector and was one of the main drivers of China's spectacular economic growth ([Naughton, 2018](#)).

Yet this liberalization also created new political risks. A growing private sector and a more active civil society increased the possibility that economic actors outside the state could demand greater autonomy, rights, and accountability ([Dickson, 2016](#); [Gold, 1990](#); [O'Brien & Li, 2006](#); [Perry, 2008](#); [Teets, 2009](#)). In response to these consequences of economic liberalization, the Party gradually reasserted control over both the economy and society, especially from the mid-2000s onward, in a pattern consistent with state power expansion described by [Way \(2005\)](#). It re-centralized authority over SOEs, strengthened their dominance in strategic sectors, relied on state-controlled financial institutions to channel resources toward state firms, and, under Xi Jinping, expanded regulatory oversight and Party presence within private firms ([Hsueh, 2011](#); [V. C. Shih, 2008](#); [Song, Storesletten, & Zilibotti, 2011](#)).⁷ In short, market liberalization was first tolerated to generate growth, but once the private sector became economically and politically consequential, the regime moved to bring it back under tighter state control.

As a critical aspect of state power expansion, the Chinese state's capacity to extract tax revenues has also significantly increased over the past two decades, particularly in the collection of VAT. Like many other countries, China adopted and expanded the VAT during periods of fiscal stress

⁷How the State Runs Business in China. The Guardian. <https://bit.ly/3T2dvko>.

faced by the central government. In 1994, the CCP Central Committee introduced a tax-sharing system (“*fen shui zhi*”) to strengthen the central government’s fiscal capacity. The most important new tax was the value-added tax (VAT) levied on most manufactured goods at a uniform rate of 17%, while other industries still paid the Business Tax (BT). The VAT was collected by a new central tax agency, the State Administration of Taxation (SAT). The SAT is a ministerial-level department under the direction of the State Council and is responsible for collecting taxes and enforcing state revenue laws. It has branch offices in most counties in China. The VAT revenues were shared between the central and local governments at a 75:25 ratio in the 1990s.

As part of the revision to the 1994 tax arrangement, the central government implemented the “VAT for business tax” reform (“*ying gai zeng*”, or VAT reform for short) to replace business taxes with the VAT from 2012 to 2016. While this reform reflected fiscal power centralization and weakened local governments’ fiscal autonomy, it also substantially expanded state power over society. The VAT is an important tax in China because it can reduce tax evasion. Like many developing countries, China relies heavily on indirect taxes on goods and services rather than individual taxpayers, as they are easier to collect. But this led to a narrower tax base and a high nominal tax rate. In China, most taxes are levied directly on enterprises, while the nominal tax rates are too high for firms to afford. Consequently, local officials give firms huge amounts of tax breaks to foster economic growth, especially before officials’ performance evaluations (L. Chen & Zhang, 2021), and tax evasion was rampant in the post-reform era (C. Zhang, 2021). Since the business tax is levied on firms’ total sales revenue without subtracting firms’ intermediate input costs, tax evasion is easy as long as firms do not issue invoices for their sales (and they have no incentives to do so). In contrast, the VAT relies on invoice-based input tax credits, creating a chain of tax enforcement that makes evasion much more difficult. Thus, the shift from the BT to the VAT can significantly expand the strength and scope of state power in tax extraction.

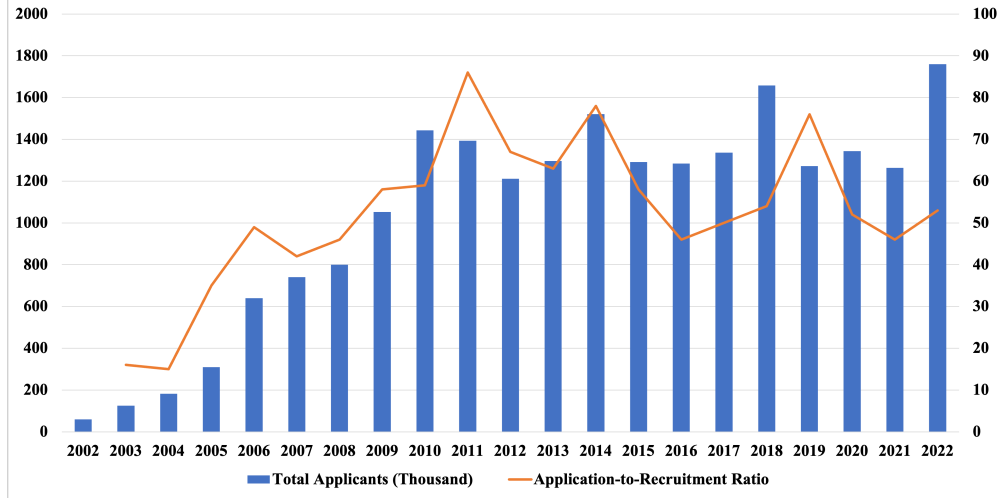
The VAT reform marked an important milestone in the expansion of taxation capacity in China. Figure A2 shows VAT revenues and total tax revenues relative to the total budgetary fiscal revenues at the national level over time. Following the reform in 2016, the share of VAT in government fiscal revenues has significantly increased. With greater reliance on VAT as a tax source since 2016, the government has seen an increase in the share of total tax revenue within the overall fiscal revenue, reversing the previous downward trend. Both revenue trends suggest an increase in

the government's tax extraction capacity following the reform. The reform directly reflects state power expansion by enhancing central fiscal capacity and expanding the scope of state power. Specifically, it strengthened the centralized State Administration of Taxation (SAT) and its local branches, the agency responsible for collecting VAT.

3.3 China's Civil Servant Fever

Alongside the power expansion of the Chinese state, the number of NCSE applicants increased sharply in the 2000s. In 2003, the number doubled to 125,000 from that of the previous year. In 2005, China enacted the Civil Servant Law, which took effect in 2006 and codified civil service recruitment based on open examinations, rigorous evaluation, and equal competition, marking the institutionalization of merit-based recruitment in the civil service (X. Huang, 2026; H. Liu, 2023). Since then, there has been a steady increase in the number of applicants and the applicant-to-recruitment ratio (Figure 1). The number of applicants reached 1.4 million in 2010 and became relatively stable afterward. In 2022, more than 2.12 million participated in the NCSE, while the total number of college graduates in that year was 9.1 million. There are also huge variations in the applicant-to-recruitment ratio. In 2022, the average number of applicants per job opening is 68. Some jobs received zero applicants, whereas the most popular job attracted 20,602 applicants. With a waning job market due to the COVID-19 pandemic, more and more individuals, including many fresh college graduates, prioritize seeking stable civil service jobs by taking the NCSE.

The NCSE is administered by the State Administration of Civil Service (SACS). It fills positions for three types of government organizations: central party organs, central state administrations (including their local branches), and public administrations governed by the Civil Servant Law. The NCSE fills positions in central departments as well as their branch offices at all levels of government, including the provincial, prefecture, county, and township levels. Parallel to the NCSE is the Provincial Civil Service Exam (PCSE), which fills positions only at local levels. Our empirical study focuses on the NCSE because it includes positions at both central and local levels, especially considering that the VAT reform directly influences the power of the State Administration of Taxation, a central tax agency with local branches all over China.



Source: Data for 2002–2009 come from the Beijing News at <https://bit.ly/4bKJCg3>. Data for 2010–2022 are authors’ calculations.

Figure 1: Total Number of Applicants and Application-to-Recruitment ratio by Year

4 Data and Empirical Strategy

This section first introduces the comprehensive database of NCSE job openings we compiled. It then outlines a difference-in-differences design that uses the VAT reform as a policy shock to causally estimate the effect of state power expansion on citizens’ preferences for civil servant jobs.

4.1 Data and Measures

Our NCSE database contains demand- and supply-side information on China’s civil service job market from 2011 to 2022. Each October, the central government releases vacancy postings for the next year’s NCSE. Applicants then select one position before taking the written exam, after which the government announces the minimum score required for each position’s interview stage.

The demand-side information in our NCSE data comes directly from the full list of job postings, which detail the central government’s needs and requirements for potential civil servant candidates. The supply side information is reflected by the number of applicants for each NCSE position every year. We also collected information about the interview cutoff score for each position as a proxy for the ability level of the candidate pool. The 2011-2014 NCSE data were collected from the government website. The 2014-2021 NCSE data were scraped from the websites of two major companies that provide training materials to NCSE takers, *Huatu Education* and *Zhonggong Education*.

The prefecture-level business tax (BT) and value-added tax (VAT) data for 2007 to 2011 are from the Regional Statistical Yearbooks 2008 to 2012 as in L. Zhang, Lu, and Liu (2022). We merge the prefecture-level tax data with the NCSE dataset based on the job location at the prefecture level and calculate the BT-to-VAT ratio in 2011. Excluding prefectures with missing information and extreme values for BT-VAT ratios, we end up with 288 prefectures.⁸ In addition, we collected a set of prefecture-level economic variables from the China City Statistical Yearbook 2011 to 2021 as our covariates.

From the NCSE database, we mainly focus on three outcome variables. The first outcome variable is the total number of applicants for an NCSE position i in year t , denoted as $Applicants_{i,t}$.⁹ To measure the competitiveness of a position, we construct the *Applicant-to-Recruitment (App-to-Recruit)* ratio of position i in year t as the ratio of the total number of applicants and the total number of recruitment slots, i.e., $App-to-Recruit_{i,t} = \frac{Applicants_{i,t}}{Recruitment_{i,t}}$. The last outcome measure is the interview cutoff score for position i in year t , denoted as $Min. Score for Interview_{i,t}$.¹⁰ Out of a total of 166,012 positions, we exclude those with missing information in the application, recruitment, locations, and agency affiliations, which reduces our sample to 151,944 positions. We further exclude observations with missing BT-VAT ratios, retaining a sample of 137,245 job openings across all hiring agencies over 12 years.

4.2 Empirical Strategy

To causally identify the effect of state power expansion on citizens' preferences for civil servant jobs in China, we take advantage of the VAT reform and exploit subnational variations in pre-reform tax compositions—the ratios of business taxes to value-added taxes (BT-VAT ratios). Figure 2 presents the geographic distribution of BT-VAT ratios across prefectures in China in 2011, the year before the pilot reform. Intuitively, the VAT reform should have a larger impact on localities where the government levied more business taxes than value-added taxes before the reform. In

⁸The tax data for 48 prefectures, primarily ethnic minority regions in Yunnan, Xinjiang, and Tibet, are not included in the yearbooks. Sanya City in Hainan Province recorded an extreme BT-VAT ratio of 23.25 in 2011 (the national average, excluding Sanya, is 2.164).

⁹ t represents the year of registration and examination, a year before the recorded NCSE year.

¹⁰This variable is only available from 2014 to 2021. To compare applicant qualities across years, we rescale the minimum score for interview into a 200-point scale for all job positions (most positions use a generic exam with a maximum score of 200 points, whereas exams for positions in law enforcement, requiring uncommon languages, or in financial regulations have a maximum score of 100 points).

particular, because the VAT makes tax evasion more difficult, tax agencies in localities with higher BT-VAT ratios have larger increases in regulatory power and can potentially collect more tax after the reform. In addition, since the SAT collected the VAT whereas the Local Administration of Taxation levied the BT before the reform, replacing the BT with the VAT enlarges the tax base for the SAT. This gives the SAT more power and revenue in localities with higher BT-VAT ratios than the SAT in localities with lower BT-VAT ratios. Given these perceived increases in power and revenue in SAT agencies, we expect the VAT reform to have a larger effect on individuals' preferences for SAT jobs in localities with higher BT-VAT ratios.

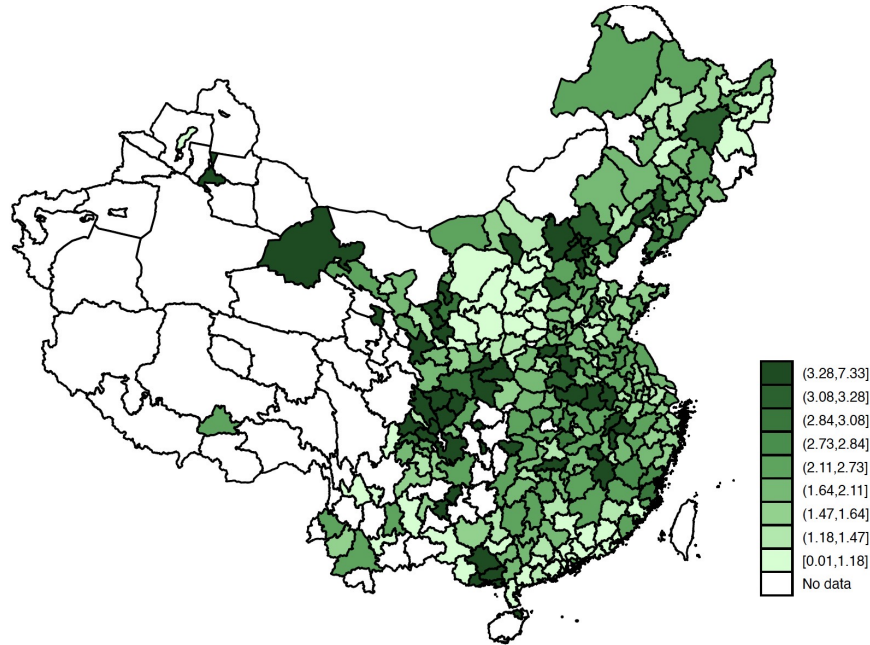


Figure 2: BT-to-VAT Ratio by Prefectures, 2011

Based on this logic, we construct a dummy variable that equals 1 if a prefecture has a pre-reform BT-VAT ratio greater than the national average and 0 otherwise.¹¹ We end up with 113 prefectures with pre-reform BT-VAT ratios above the national average and 175 prefectures with pre-reform BT-VAT ratios below the national average. Note that, following [Cao, Xu, and Zhang \(2022\)](#), we did not use the BT-VAT ratio as a continuous treatment measure for two reasons. First, prefectures' BT-VAT ratios are likely left-skewed because few prefectures have a well-developed tertiary industry—the sector previously subject to the BT levy. Second, individuals' responses to the VAT reform are unlikely to be *linear* with respect to the reform's intensity in different localities:

¹¹When constructing this dummy variable, we do not include extreme values of the BT-VAT ratio.

they tend to remain insensitive to small to moderate changes in tax composition until a certain threshold is reached. That said, robustness checks using the continuous measure in Appendix Table A3 yield results consistent with the dichotomous specification.

We divide our sample into two subsamples: the SAT sample, which includes jobs affiliated with the State Administration of Taxation, and the non-SAT sample, which comprises jobs affiliated with all other state agencies. We expect that the VAT reform affects people’s preferences for SAT jobs rather than non-SAT jobs. Table A1 presents the summary statistics of the treatment and outcome variables by the two subsamples.

We designate post-2016 as the treatment period because the VAT reform was not fully implemented until mid-2016. See Table A2 for the timeline of the roll-out of the reform. The pre-2016 pilot reform in transportation and related sectors only affected a small fraction of the economy. As shown in Figure A3, the share of industrial output from the transportation, storage, and postal-service sectors accounted for roughly 4% of the total GDP. Given the very small share of these sectors in the economy, the pilot reform should have a negligible impact on individuals’ job preferences.¹² Therefore, we define our post-treatment year starting from 2017 and estimate the following model:

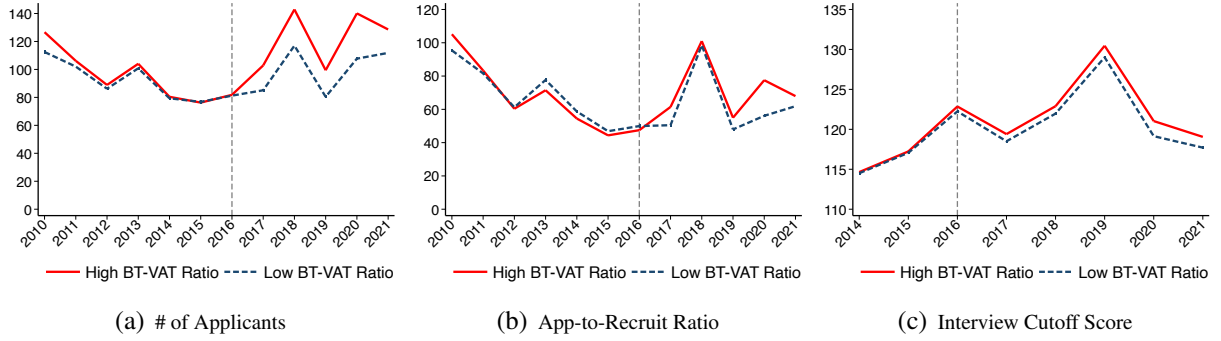
$$Y_{i,t} = \alpha_0 + \beta (\text{High BT-VAT}_p \times \text{Post2016}_t) + \alpha_1 X'_{i,t} + \alpha_2 P'_{p,t} + \delta_p + \delta_t + \epsilon_{i,t}, \quad (1)$$

where $Y_{i,t}$ is one of the three outcome variables: the total number of applicants per position, the App-to-Recruit ratio, and the interview cutoff score. High BT-VAT_p is a dummy variable that equals 1 if a position is located in a prefecture with a high BT-VAT ratio before the reform (in 2011), and zero otherwise. Post2016_t is a dummy variable that equals 1 if the calendar year is after 2016. δ_p and δ_t are prefecture and year fixed effects, respectively. $X_{i,t}$ is a set of position-specific characteristics, and $P'_{p,t}$ is a set of prefecture-level socioeconomic factors. We particularly include the combined share of the secondary and tertiary sectors in total GDP to account for the possibility that prefectures with high BT-to-VAT ratios were those where service industries—such as finance, law, and IT—made up a larger share of economic output prior to the reform. These sectors are typically labor-intensive and benefit less from input-based VAT deductions. See Online

¹²We also control for the pilot reform in the robustness check and find minimal effects. See Table A5 for details.

Appendix A1 for detailed information about the covariates.

We are interested in β , which captures the change in individuals' preferences for civil servant jobs located in the high BT-VAT regions versus low BT-VAT regions following the VAT reform. The parallel pre-trends observed between the two groups prior to 2016 (Figure 3) support the validity of the DiD estimation. A placebo test using a dynamic DiD model further shows the lack of pretreatment effects (see Figure 4 in Section 5.2).



Notes: Annual averages by high versus low BT-VAT groups are plotted using the SAT sample.

Figure 3: Average NCSE Outcomes by High and Low BT-VAT Prefectures

5 Empirical Findings

This section presents the main results on the effects of the VAT reform on talent attraction, conducts a series of robustness checks, and examines the underlying mechanisms. It also explores the sources from which the central state attracts talent.

5.1 Main Results

Table 1 Panel A reports the main effects of the VAT reform on talent recruitment for civil servant jobs affiliated with the State Administration of Taxation (the SAT sample). Columns (1), (3), (5) include prefecture and year fixed effects. In Columns (2), (4), and (6), we add prefecture-level and position-level covariates. Consistent with our expectations, job openings in prefectures with high pre-reform BT-VAT ratios after the VAT reform see increases in the number of applicants (H1), the App-to-Recruit ratio (H2), and the minimum scores required for interviews (H3) compared with job openings in prefectures with low pre-reform BT-VAT ratios. The magnitudes

are large. We observe an increase of 20 additional applicants (Column [2]) and an 11-percentage-point rise in the Applicant-to-Recruit ratio (Column [4]) for each job opening in high BT-VAT ratio prefectures compared to those in low BT-VAT ratio prefectures after the reform. Both represent increases of more than 20% relative to their pre-reform means. Column (6) further indicates that the minimum score for interviews increased by 1.3 points for each job opening in high BT-VAT ratio prefectures compared with those in low BT-VAT ratio prefectures (the mean is 120 points). This is a significant increase because, in a standard exam, a one-point difference can eliminate a large number of contestants. The findings suggest that the VAT reform increased the number of applicants, made jobs more competitive, and improved the ability level of the recruited talent for jobs affiliated with the State Administration of Taxation and its local branches.

By contrast, we do not expect the VAT reform to have a significant impact on the non-SAT sample, as the reform does not alter the powers of those non-SAT government agencies. Indeed, Table 1 Panel B shows that the reform has no statistically significant effects on the popularity and competitiveness of non-SAT jobs. After controlling for prefecture and position covariates, the effect of the reform on the minimum score for interviews in the non-SAT sample is only significant at the 10% level and is half of the magnitude observed in the SAT sample. These results are not only consistent with our expectations but also suggest that the talent attracted to SAT-affiliated departments is unlikely to be drawn from the talent pool of other state departments.

5.2 Long-term Effects

We use a dynamic DiD model to allow for treatment effects to change over time. Figure 4 shows that the treatment effects on the number of applicants, the App-to-Recruit ratio, and the minimum scores required for interview eligibility in the SAT sample remain substantial and statistically significant even five years after the VAT reform. This suggests that the reform increased the popularity of SAT jobs in the long term.

5.3 Robustness Checks and Alternative Explanations

In the main analysis, we divide the sample into two groups by the mean value of the pre-reform BT-VAT ratio at the prefecture level. This approach is used because individuals' perceptions of the VAT reform are unlikely to be linear; they may remain insensitive to small or moderate changes in tax composition. Nevertheless, we replicate the analysis using the continuous measure of the

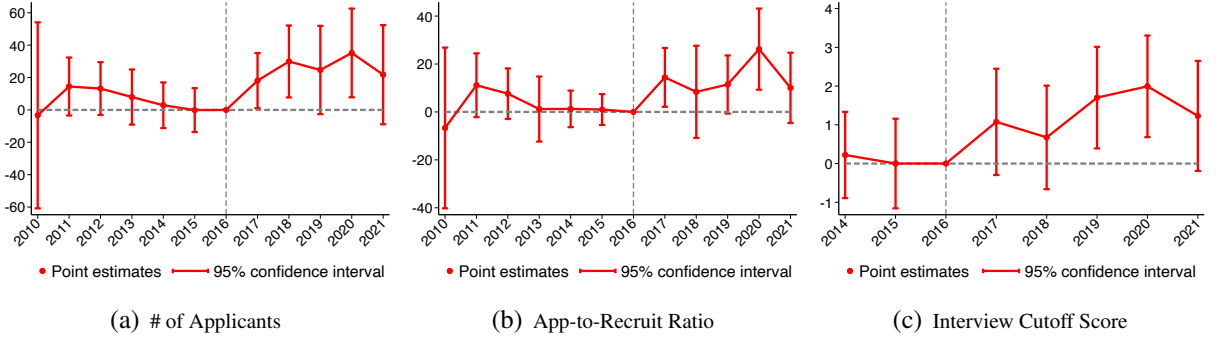
Table 1: Effects of the VAT Reform on NCSE Outcomes

Panel A (SAT Sample, Main Analysis)	# of Applicants		App-to-Recruit Ratio		Interview Cutoff Score	
	(1)	(2)	(3)	(4)	(5)	(6)
D.V. Mean	98.56	97.07	63.12	61.29	120.5	120.5
High BT-VAT × Post2016	18.943* (9.627)	20.050** (9.252)	10.895** (4.793)	11.271** (4.548)	1.098*** (0.397)	1.310*** (0.401)
Observations	75,498	70,498	75,498	70,498	56,892	56,086
R^2	0.068	0.104	0.059	0.094	0.357	0.422
Prefecture FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Prefecture Covariates	No	Yes		Yes	No	Yes
Position Covariates	No	Yes	No	Yes	No	Yes
Panel B (Non-SAT Sample, Placebo)	# of Applicants		App-to-Recruit Ratio		Interview Cutoff Score	
	(1)	(2)	(3)	(4)	(5)	(6)
D.V. Mean	122.9	119.5	85.12	83.42	119.4	119.5
High BT-VAT × Post2016	-11.183 (10.975)	-6.454 (10.208)	-6.735 (6.265)	-5.451 (4.206)	1.303*** (0.395)	0.707* (0.371)
Observations	61,747	56,502	61,747	56,502	41,794	40,244
R^2	0.043	0.095	0.038	0.107	0.225	0.338
Prefecture FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Prefecture Covariates	No	Yes	No	Yes	No	Yes
Position Covariates	No	Yes	No	Yes	No	Yes
Number of Provinces	31	31	31	31	31	31
Number of Prefectures	288	288	288	288	288	288

Notes: Prefecture-level and position-level covariates are added in Columns (2), (4), and (6). Standard errors reported in parentheses are clustered at the prefecture level: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

pre-reform BT-VAT ratio (Appendix Table A3). The results are consistent with those from Table 1. We also use the 2007-2011 (pre-reform period) average BT-VAT ratio instead of the 2011 value to divide the sample, and the results remain robust (Appendix Table A4).

One may also be concerned that our results could be driven by an increase in government job openings in prefectures with high BT-VAT ratios after the reform rather than reflecting changes in applicants' preferences. For example, many local-tax bureau cadres would probably be transferred to the SAT following the reform. Thus, fewer positions would be available for new recruitment, resulting in more intense competition.



Note: Point estimates and 95% confidence intervals are estimated based on the SAT sample.

Figure 4: Dynamic Effects on NCSE Outcomes

We address this concern by looking at the impact of the VAT reform on SAT agencies' recruitment quota directly. Results are presented in Table 2. In Column (1), we examine the impact of the VAT reform on the recruitment quota per position. We find no significant change in high BT-VAT ratio prefectures relative to the low BT-VAT regions. In Column (2), we aggregate the position-level recruitment quotas across all SAT positions for each prefecture and create a prefecture-year panel. We find that the total number of openings did not change significantly in high BT-VAT regions relative to low BT-VAT regions after the reform. These results suggest that the patterns we find in Table 1 are unlikely to be driven by an increase in job openings.

Table 2: Effects of the VAT Reform on NCSE Position-Level Demand and Prefecture-Level Aggregate Demand

SAT Sample	Recruitment Quota (position-level)	Aggregate Recruitment Quota (prefecture-level)
	(1)	(2)
D.V. Mean	1.774	38.99
High BT-VAT $\times$ Post2016	-0.041 (0.061)	2.411 (2.111)
Observations	70,867	3,385
R^2	0.260	0.776
Prefecture FEs	Yes	Yes
Year FEs	Yes	Yes
Prefecture Covariates	Yes	Yes
Position Covariates	Yes	
Number of Provinces	31	31
Number of Prefectures	288	288

Notes: Standard errors reported in parentheses are clustered at the prefecture level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We further conduct a series of robustness checks to rule out alternative explanations for our findings. First, our results are robust even after controlling for province-specific heterogeneous trends and time-varying effects of pre-pilot economic conditions in 2011 (Table A6). Second, we show that the results are unlikely to be driven by the 2019 integration of Local Tax Administrations into the SAT, as the integration was proposed only in late 2018, took effect in 2019, and would be expected to reduce—rather than increase—the attractiveness of SAT positions due to the crowded agency after the integration. Against this explanation, our dynamic estimates reveal significant increases in applications and competition as early as 2017 (Figure 4), and the results remain robust when restricting the sample to pre-2019 years (Table A7), thereby also alleviating concerns about confounding shocks from both the integration and the COVID-19 pandemic. Third, we demonstrate that exam takers’ anticipatory behavior around the reform does not bias our estimates by excluding the 2016–2017 transition period (Table A8). Finally, we implement a triple-difference design that exploits agency-by-prefecture-by-year variations to show that the post-2016 increase in applications and competition is specific to SAT agencies in high BT-VAT prefectures (Table A9), confirming that the VAT reform—rather than broader labor-market or institutional trends—drove the surge in talent attraction.

5.4 Mechanisms: Power, Benefits, and Graft Opportunities

To test the mechanisms through which the VAT reform made SAT jobs more attractive, we first conduct an online survey of 3,700 individuals from 28 provinces in China to identify NCSE takers and ask directly about their perceptions of the VAT reform. Online Appendix A4 provides details about this survey. Table 3 presents descriptive evidence. Out of the total 2,059 respondents who are potentially eligible for the NCSE in terms of their educational attainment, around 36% (747 individuals) had taken or will take the NCSE. Within this group, a substantial proportion, about 71% of respondents (529 individuals), had applied or will apply for positions affiliated with the SAT. Among them, awareness of the VAT reform was high, with 87% (459 individuals) indicating familiarity with the reform. Consistent with our argument, a majority (63%, 286 individuals) of those aware of the VAT reform believed that the reform increased the power of the SAT department. Additionally, about 71% (316 individuals) perceived that after the VAT reform, government officials working in the SAT experienced an increase in compensation and welfare.

Table 3: Individual Perceptions of Government Jobs in Response to the VAT Reform

Take NCSE?	Apply SAT?	Know VAT Reform?	Power?	Compensation & Welfare?
Yes (747/36.28%)	Yes (529/70.91%)	Yes (459/86.77%)	Increase (286/62.58%)	Increase (316/70.54%)
			No Change (80/17.51%)	No Change (81/18.08%)
			Decrease (91/19.91%)	Decrease (51/11.38%)
	No (217/29.09%)	No (70/13.23%)		
No (1312/63.72%)				

Notes: The sample includes individuals who are eligible for the NCSE in terms of their educational attainment. See Appendix A4 for details about the survey items.

Moreover, Table A10 presents a regression analysis based on the survey data. Results show that elites—individuals in higher social classes or with higher educational attainment—are more likely to perceive increases in power and benefits associated with SAT agencies after the VAT reform. In particular, elites are significantly more likely to expect an increase in power following the VAT reform. Together, the two pieces of evidence from this survey support our argument that job applicants, especially those in higher social classes or with higher education levels, applying for positions affiliated with the SAT in the NCSE perceive that the reform has made those positions more powerful and lucrative.¹³

As discussed in the theory section, state power expansion in non-democratic regimes makes government jobs particularly attractive because the lack of power constraints on government officials allows them to benefit from the amplified power and regulatory prowess. This logic implies that powerful positions in the state sector should attract more applicants than positions with less power. To test this hypothesis (H4), we examine the heterogeneous effects of the VAT reform on the popularity of different job openings within the SAT agencies. Certain roles in the SAT agencies entail greater discretionary power over taxation issues of the private sector, endowing SAT employees with more authority, privileges, and graft opportunities. Our interviews with SAT employees indicate that roles in tax-base management, auditing, and enforcement possess greater authority than other jobs, such as front desk and office jobs. If the VAT reform amplified the power and benefits linked to certain SAT roles, job posts involving such roles would be perceived by applicants as particularly empowering and therefore attract more applicants.

¹³Figure A6 plots the conditional probabilities of perceiving an increase in power and in compensation and welfare. Conditional means are estimated at each level of individual education and social status using a local polynomial smoothing approach.

We utilize the description and title of job posts in the NCSE dataset to identify powerful positions. Following the suggestions of our interviewees working in tax agencies, we consider a position “powerful” if the textual information contains keywords such as “tax base,” “tax inspection,” “tax audit,” and “tax law enforcement.” In total, we identified around 12% of all SAT positions as “powerful.” Note that “powerful” and “less powerful” positions are in relative terms. Appendix A5 provides detailed information about the categorization.

Table 4: Effects of the VAT Reform on NCSE Outcomes: Heterogeneous Analysis

Variables	# of Applicants		App-to-Recruit		Interview Cutoff Score	
	(1)	(2)	(3)	(4)	(5)	(6)
Positions	Powerful	Less powerful	Powerful	Less powerful	Powerful	Less powerful
D.V. Mean	98.37	96.93	64.02	60.96	121.5	120.4
High BT-VAT $\times$ Post2016	31.713* (16.869)	17.751* (9.451)	21.274** (8.838)	9.381** (4.695)	0.278 (0.790)	1.381*** (0.405)
N of Observations	7,723	62,757	7,723	62,757	4,636	51,431
R^2	0.157	0.109	0.140	0.097	0.499	0.423
Prefecture FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Prefecture Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Position Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Number of Provinces	29	31	29	31	28	31
Number of Prefectures	190	288	190	288	151	288

Notes: Columns (1), (3), and (5) are for powerful SAT-agency positions, while Columns (2), (4), and (6) are for less powerful SAT-agency positions. Standard errors reported in parentheses are clustered at the prefecture level: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Please refer to Appendix A5 for the details of categorizing powerful and less powerful SAT positions.

Table 4 presents the results for both “powerful” and “less powerful” jobs. We observe an 80% larger increase in the number of applicants and a 130% larger increase in the applicant-to-recruitment ratio for “powerful” positions, compared to “less powerful” positions. This evidence suggests that the VAT reform made powerful SAT positions much more attractive to applicants than less powerful ones. Additional analyses in Table A12 show that this pattern is largely driven by positions involving tax inspection, or “*ji cha*” (稽查), which have the clearest coercive and discretionary authority within the SAT. This is consistent with our argument that applicants are especially drawn to state positions where expanded regulatory power is most directly exercised. We find that the increase in the minimum score required for interview eligibility for “powerful”

positions is statistically insignificant. This is reasonable, as “powerful” positions have already been attracting high-ability applicants even before the VAT reform, as indicated by their higher pre-reform average minimum score (120 points for powerful positions versus 117 points for less powerful positions). After the reform, the average minimum score for “powerful” positions (120.6) is still higher than that for “less powerful” positions (119.5). Nevertheless, the heterogeneous effects of the VAT reform among different positions provide strong evidence for the mechanism that state power expansion makes government jobs more attractive by increasing the regulatory power of those jobs.

We also provide suggestive evidence that candidates may prefer positions with greater graft opportunities by examining how the effect of the VAT reform varies with regional differences in historical corruption levels. Using the anti-corruption campaign database from [Y. Wang and Dickson \(2022\)](#), we construct a prefecture-level measure of corruption based on the number of past cases. We find that, following the reform, SAT positions in regions with higher perceived graft opportunities attracted significantly more and higher-ability applicants (Table [A13](#)).

5.5 State-Society Talent Allocation

Our main results show that the VAT reform attracted more talent to positions in the State Administration of Taxation (SAT), and that this talent did not come from the pool of applicants for non-SAT jobs within the national civil service system. However, it remains possible that these applicants were not drawn from broader society but instead consisted of individuals who would otherwise have applied for positions in the Local Administrations of Taxation (LATs), especially given that the VAT reform also entailed fiscal power centralization.

To address this concern, we examine the effect of the VAT reform on applications to the Local Administrations of Taxation (LATs) using data from multiple years of the Provincial Civil Service Exams (PCSE) conducted in Zhejiang Province.¹⁴ Table [5](#) compares the effects of the VAT reform on LAT job applications (PCSE) and SAT job applications (NCSE) in Zhejiang Province, using the same model specification and treatment variable as in our main analysis.¹⁵

¹⁴The PCSE is administered by each province and is responsible for recruiting positions for the LATs. The data are obtained directly from the Zhejiang Provincial Civil Service Examination and Recruitment Website. Table [A11](#) presents summary statistics for the PCSE sample.

¹⁵Table [A11](#) presents the summary statistics of the treatment and outcome variables for the PCSE dataset from Zhejiang Province.

We find that following the VAT reform, both the number of applicants (Columns [1] and [2]) and the application-to-recruitment ratio (Columns [3] and [4]) increased for local and national tax agencies in Zhejiang.¹⁶ Panel A, Columns (5) and (6), further show that the VAT reform did not reduce recruitment quotas for local taxation positions. Although this analysis is limited to Zhejiang Province due to data availability constraints, the positive effects observed for both PCSE and NCSE in the same province suggest that the VAT reform is unlikely to reflect a reallocation of government talent from local tax agencies to the central tax authority.

Table 5: The Effect of the VAT Reform on Local Administration of Taxation in Zhejiang Province

Panel A: PCSE local tax agency, Zhejiang Province						
	# of Applicants		App-to-Recruit Ratio		Recruitment Quota	
	(1)	(2)	(3)	(4)	(5)	(6)
D.V. Mean	67.66	67.66	35.26	35.26	1.764	1.764
High BT-VAT $\times$ Post2016	52.525*** (10.709)	48.862*** (14.482)	0.951 (2.782)	25.688 (15.995)	0.433*** (0.115)	-0.336 (0.440)
Observations	1,078	1,078	1,078	1,078	1,078	1,078
R^2	0.152	0.153	0.101	0.105	0.127	0.132
Panel B: NCSE SAT, Zhejiang Province						
	# of Applicants		App-to-Recruit Ratio		Interview Cutoff Score	
	(1)	(2)	(3)	(4)	(5)	(6)
D.V. Mean	108.5	107.8	73.96	73	125.6	125.6
High BT-VAT $\times$ Post2016	54.650*** (12.588)	43.142*** (9.213)	28.204** (12.543)	22.186* (11.637)	1.520* (0.743)	1.764 (1.533)
Observations	4,288	4,066	4,288	4,066	3,276	3,276
R^2	0.046	0.117	0.043	0.122	0.191	0.298
Controls						
Prefecture FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Pref. Covar.		Yes		Yes		Yes
Position Covariates		Yes		Yes		Yes
Number of Provinces	1	1	1	1	1	1
Number of Prefectures	11	11	11	11	11	11

Notes: In panel A, we collect data on applications, app-to-recruit ratios, and recruitment quotas for Local Administration of Tax (LAT) positions from the Provincial Civil Service Exam (PCSE, for 2014, 2015, 2016, 2018) in Zhejiang Province. For comparison, in Panel B we use our main NCSE dataset but restrict the sample to SAT positions from Zhejiang province only.

To explore the impact of the VAT reform on the general labor market, we use six waves of the Chinese General Social Survey (CGSS) from 2010 to 2017.¹⁷ The CGSS is a nationally represen-

¹⁶The dynamic estimates (Figure A4) show no evidence of pre-trends in the Zhejiang LAT data.

¹⁷We use the CGSS data because data on China's private sector labor market are not available. The CGSS data

tative survey project run by Chinese academic institutions since 2003. It contains a wide range of employment information, including the ownership type of an individual’s employer. We aggregate our prefecture-level BT-VAT measure to the provincial level and merge it with the CGSS data, as the province is the finest geographic unit in the CGSS.¹⁸ The average BT-VAT ratio at the provincial level is about 2.4, which is comparable to the prefecture-level average. Following our main analysis, we construct a dummy variable, “High BT-VAT”, which equals 1 if a province’s 2011 BT-VAT ratio is above the national mean, and 0 otherwise.

Table 6: The Effect of the VAT Reform on Private Sector Employment, CGSS

Sample: CGSS	Work in the Private Sector (as opposed to the Public Sector)					
	NCSE-eligible (College or above, and age 18-35)			Placebo: NCSE-non-eligible		
	(1)	(2)	(3)	(4)	(5)	(6)
D.V. Mean	0.530	0.530	0.530	0.703	0.703	0.703
High BT-VAT $\times$ Post 2016	-0.074* (0.040)	-0.056** (0.024)	-0.061** (0.026)	-0.021 (0.025)	-0.032 (0.023)	-0.052 (0.043)
Observations	3,893	3,805	3,805	18,691	18,611	18,611
R^2	0.100	0.214	0.219	0.210	0.259	0.261
Province FEs	Yes			Yes		
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
LnGDP	Yes	Yes	Yes	Yes	Yes	Yes
Ind. Charac.	Yes	Yes	Yes	Yes	Yes	Yes
(Province $\times$ Cohort) FEs		Yes	Yes		Yes	Yes
Province-specific time-trend			Yes			Yes
Number of Provinces	28	28	28	28	28	28

Note: The data come from the 2010, 2011, 2012, 2013, 2015, and 2017 waves of the Chinese General Social Survey (CGSS). Columns (1)–(3) use respondents who are eligible for the NCSE, defined as having a college education or above and being between ages 18 and 35. Columns (4)–(6) use respondents who do not meet either eligibility criterion and serve as a placebo group. Cohorts are defined by birth year. All specifications control for individual characteristics (gender, CCP membership, ethnicity, and urban residence) and province-level log GDP. Columns (1) and (4) include province and year fixed effects; Columns (2) and (5) include province-by-cohort fixed effects; and Columns (3) and (6) additionally include province-specific linear time trends. Our preferred specification is shown in Columns (2) and (5). Standard errors clustered at the province level are presented in parentheses: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

after 2017 has not yet been published when we were drafting this paper. During the sample period, the CGSS was not implemented in 2014 and 2016. We exclude data from Xinjiang and Tibet due to small size (85 obs and 204 obs across six waves respectively).

¹⁸Following our main analysis, we exclude Hainan Province because of its extremely high BT-VAT ratio.

We use a DiD specification to estimate the effect of the VAT reform on the general labor-market outcome. The outcome variable is a dummy variable that equals 1 if an individual works in the private sector and 0 if the individual works in the public sector.¹⁹ Within survey respondents who have active employment status, age, birth-year, and education-attainment information in non-agricultural sectors (23,137 observations in 28 provinces), we focus on individuals eligible for the NCSE, i.e., with college education and aged between 18 and 35, according to the NCSE’s requirements for education and age. Consequently, the subgroup with a high school education or below, or age beyond the above range serves as a placebo group. The VAT reform should have no effect on the placebo group.

Table 6 presents the results. Columns (1), (2), and (3) show that the VAT reform significantly reduced the likelihood that NCSE-eligible individuals worked in the private sector rather than the public sector. Disaggregating by public-sector subcategories, these individuals became more likely to work in the “party-and-government” system (*dang zheng ji guan*) and in state-owned enterprises (SOEs) (see Table A15). In contrast, Columns (4), (5), and (6) of Table 6 show no significant effects for the placebo group. The estimated magnitudes are also substantially smaller for the placebo group.²⁰

Taken together with our earlier findings that the additional talent did not come from applicant pools for either local tax agencies or non-tax central agencies, these results suggest that the expansion of the Chinese state’s tax extraction capacity redirected talent away from the private sector and toward the state sector. This shift could be attributed to both the increased power and benefits within the state sector and a decline in the private sector due to state power expansion, although it is challenging to separate these effects.

6 Conclusion

This paper distinguishes state power expansion in non-democracies from democratic state building and examines its impact on state-society talent allocation. It highlights a crucial mechanism by which state building attracts talent to government positions: state employees can benefit from the amplified regulatory power due to the lack of constraints on government officials in such

¹⁹Individuals who work in the private sector include those with their own business, or those who work in an enterprise with domestic employers, employers from Hong Kong, Macau, or Taiwan, or foreign employers.

²⁰Figure A7 shows that the results are not sensitive to the exclusion of any single survey year.

settings. Our finding that state power expansion through the VAT reform attracted more and higher-ability talent to take the NCSE in China is particularly notable against the backdrop of an ongoing corruption crackdown that deters applications for government positions (Lai & Li, 2024). Theoretically, this attraction to power and graft opportunities is more pronounced in non-democratic regimes than in democracies, which potentially explains the high appeal of civil servant jobs in China compared to the U.S. This may also help explain Boittin et al. (2016)’s empirical finding that Chinese civil servants exhibit a higher degree of meritocracy than U.S. federal employees. Future research could explore whether state power expansion has similar effects on talent allocation in more democratic contexts.

Our theory and findings have important implications for corruption and economic performance. We argue that talent is attracted to powerful state positions because unconstrained state power creates material benefits and graft opportunities. When combined with merit-based recruitment, this dynamic implies that smart but potentially dishonest individuals are more likely to self-select into public-sector jobs. This logic helps explain the findings of a recent working paper by Z. Liu, Peng, and Wang (2026), which shows that dishonest individuals, measured by graduate dissertation plagiarism, are more likely to enter and advance within China’s public sector. When such individuals fill and rise within government positions, it is unsurprising that corruption becomes deeply entrenched in the bureaucracy. Unconstrained regulatory authority along with dishonest officials could lead a country’s economy to operate through informal practices, cronyism, and political connections (Z. Chen et al., 2021; E. H. Wang, 2022). Although economic growth remains possible because petty corruption is constrained by the party-state while “access money” shapes business–government relations to facilitate growth (Ang, 2020), this model falls short of a first-best institutional arrangement.

Although talent attraction is likely an unintended consequence of state power expansion, it has important implications for political stability under unconstrained state power. It highlights a fundamental tension between political stability and economic development. In resource-poor countries such as China, economic liberalization and private-sector growth are important because sustained growth provides performance legitimacy for the regime (Dickson, 2016; Magaloni, 2006). Indeed, the state’s retreat from direct control over the private economy was a major driver of China’s growth during the reform era (Naughton, 2018). Yet a thriving private sector may also generate

demands for greater political autonomy and freedom, thereby threatening regime survival. In response, the regime may seek to reassert control by expanding state power and constraining the private sector. This process can also redirect talent from the private sector into the state, further weakening private-sector growth (Bai et al., 2025). As a result, even if expansive state power can help a country achieve rapid growth in the short run, the long-term economic prospects may be more uncertain, forcing the state to rely on other sources of legitimacy over time.

In light of the above discussion, although this paper does not examine the relationship between talent attraction and regime stability, it implies a potential mechanism of regime survival: rulers may enhance state power to attract talent to the regime, instead of leaving them in society as sources of instability (Rosenfeld, 2017). Furthermore, our argument also suggests the possibility of a feedback loop between talent allocation and state capacity. If state building along with merit-based recruitment attracts talent to the state sector, those individuals may in turn further strengthen the state's capacity and reach. This may help explain why strong states emerged in other non-democratic regimes, such as Vietnam, Singapore, pre-war Japan, and South Korea before 1988. Future research in these directions is worth exploring.

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The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Supplemental Material

Supplemental material for this article is available online.

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Online Appendix

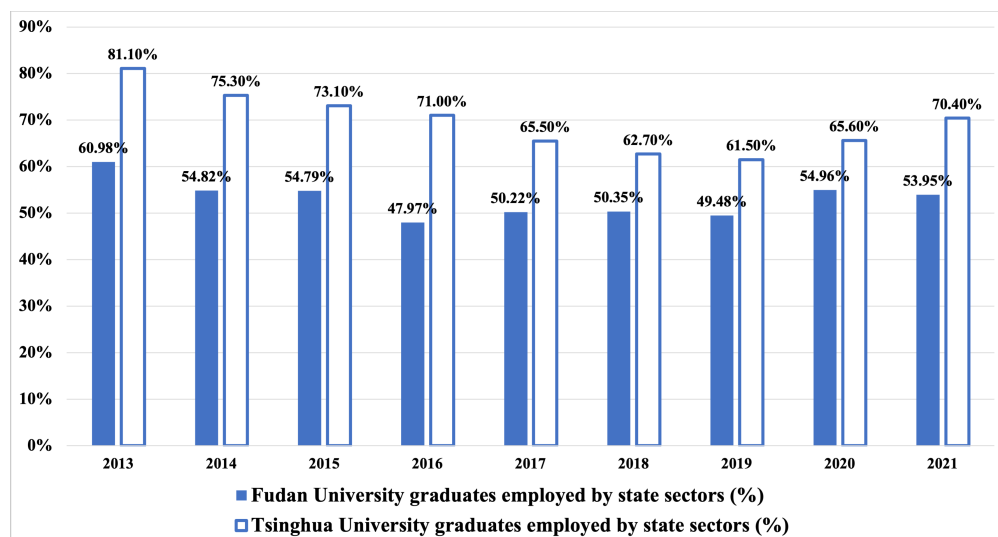
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A1 Covariates in the Constructed NCSE Data

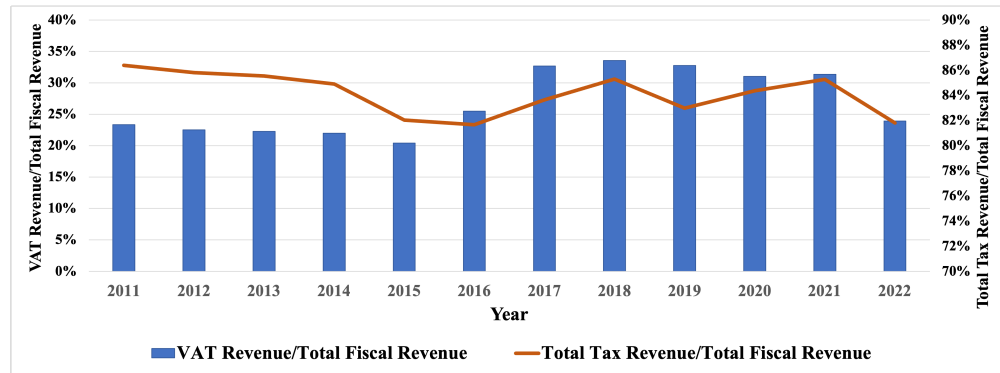
We include a set of position-specific characteristics and a set of prefecture-level covariates in the main empirical model. Position-level covariates include dummy variables for position types (regular positions or special positions such as those with special language requirements), location of jobs (whether the position is in remote or poor areas), nature of the hiring agencies (e.g., the Party organs, State Council administrative units, functional departments of the government etc.), the administrative rank of the hiring agency’s director, and requirements on education, majors, and gender. We consider these position-level covariates because job characteristics and requirements affect individuals’ preferences for the position and their eligibility to apply. Prefecture-level covariates include logged GDP per capita, logged population of permanent residents, and the combined share of the secondary and tertiary sectors’ outputs in total GDP.

A2 Appendix Figures



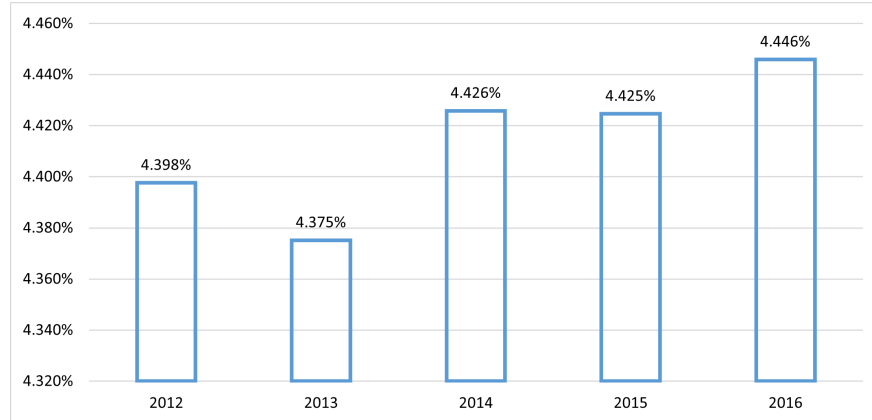
Notes: Data on state-sector employment proportions for Fudan University and Tsinghua University graduates are from <https://xxgk.fudan.edu.cn/bysjyzlndbg/list.htm> and <https://career.tsinghua.edu.cn/tzgg.htm>, respectively. State sectors include: (1) party and government system (“党政机关”); (2) public institutions (“事业单位”, including scientific R&D institutions, medical and public health agencies, schools, and others); (3) state-owned enterprises (“国有企业”); (4) the military (“部队”).

Figure A1: State Sector Employment (%) among Fudan and Tsinghua University Graduates



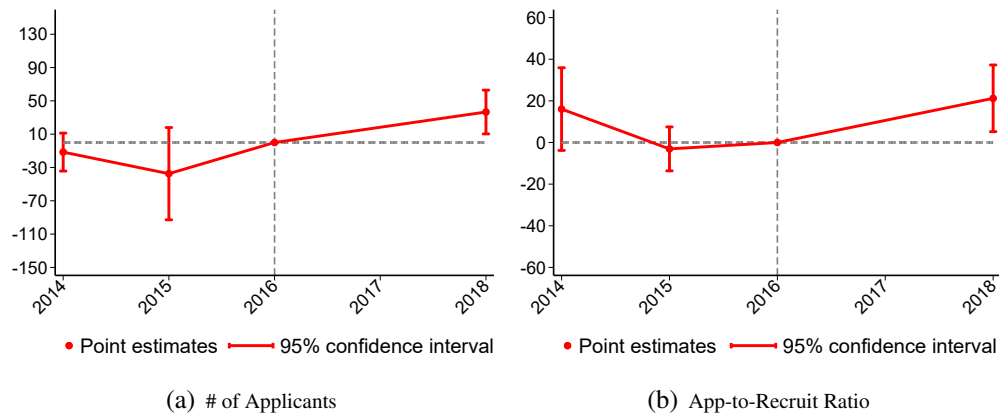
Notes: Data on VAT revenue, total tax revenue, and total fiscal revenue are from the China Statistical Yearbook (2012-2022), compiled by the National Bureau of Statistics of China. The y-axis on the left-hand side indicates the level of VAT/Total Fiscal Revenue, and the y-axis on the right-hand side indicates the level of Total Tax/Total Fiscal Revenue.

Figure A2: VAT and Total Tax Revenues as Shares of the National Total Fiscal Revenue



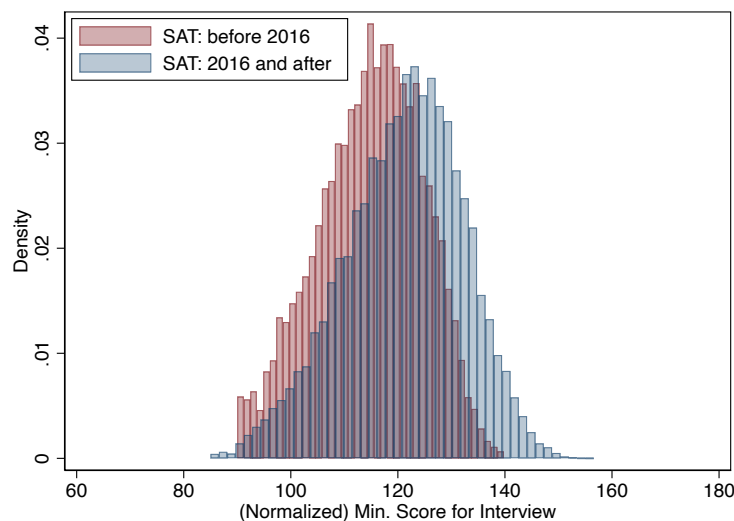
Notes: Data on GDP and sector outputs are from the China Statistical Yearbook (2013-2017), compiled by the National Bureau of Statistics of China.

Figure A3: The GDP Shares of Transportation, Storage, and Postal Industries Outputs

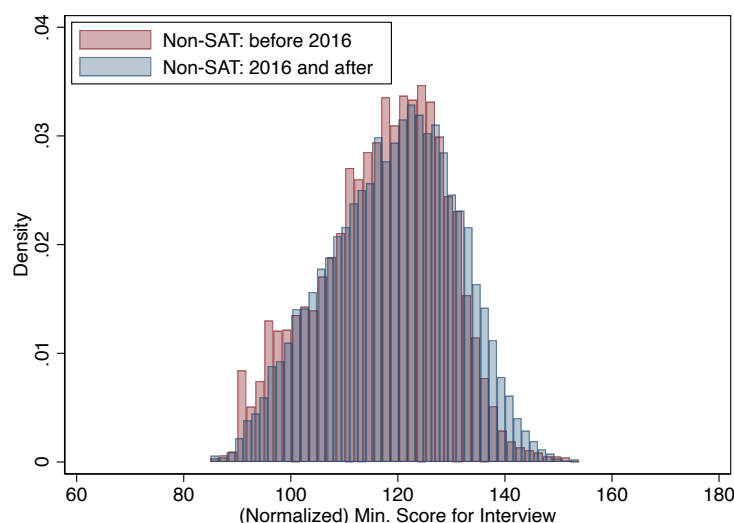


Note: Point estimates and 95% confidence intervals are estimated based on the SAT sample.

Figure A4: Dynamic Effects on PCSE Outcomes



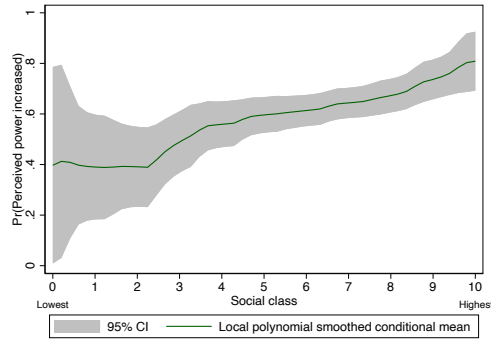
(a) The Distribution of Min. Entry Score for Interview, **SAT Sample**, before and after 2016



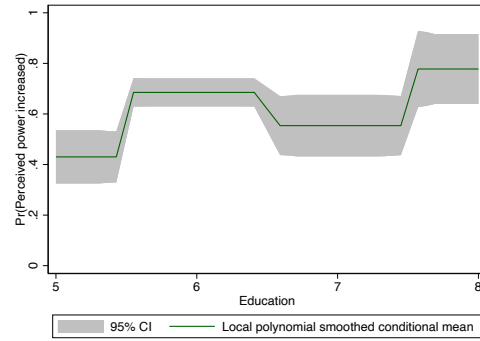
(b) The Distribution of Min. Entry Score for Interview, **Non-SAT Sample**, before and after 2016

Notes: This figure compares the distributions of the normalized minimum entry score required for interview before and after 2016, by SAT (Subfigure (a)) and non-SAT samples (Subfigure (b)). In each subfigure, the sample is split into two groups based on the timing of the VAT reform: observations from years before 2016 (represented by bins with red boxes) and those from 2016 and after (represented by bins with blue boxes). The histograms present densities, instead of frequencies, to visualize the comparison. All the scores are normalized to a 200-point scale, to be comparable across different types of NCSE job posts.

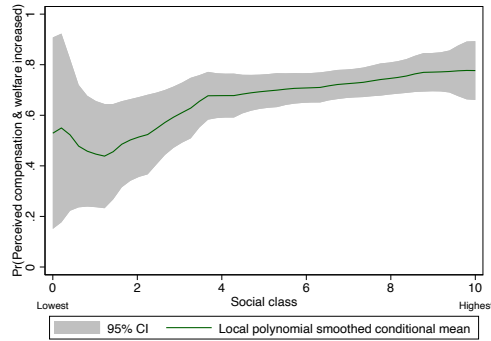
Figure A5: The Distributions of the Minimum Entry Score for Interview, before and after the VAT reform (2016), by SAT and Non-SAT Samples



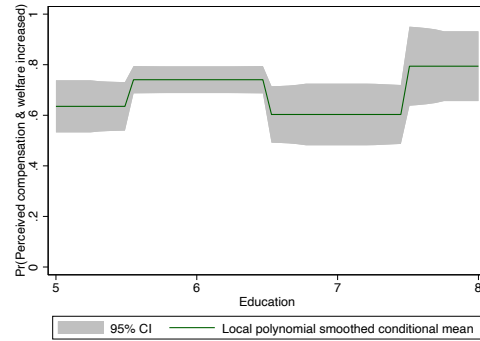
(a) Perception of Increased State Power associated with the VAT Reform, by Social Status



(b) Perception of Increased State Power associated with the VAT Reform, by Education Level



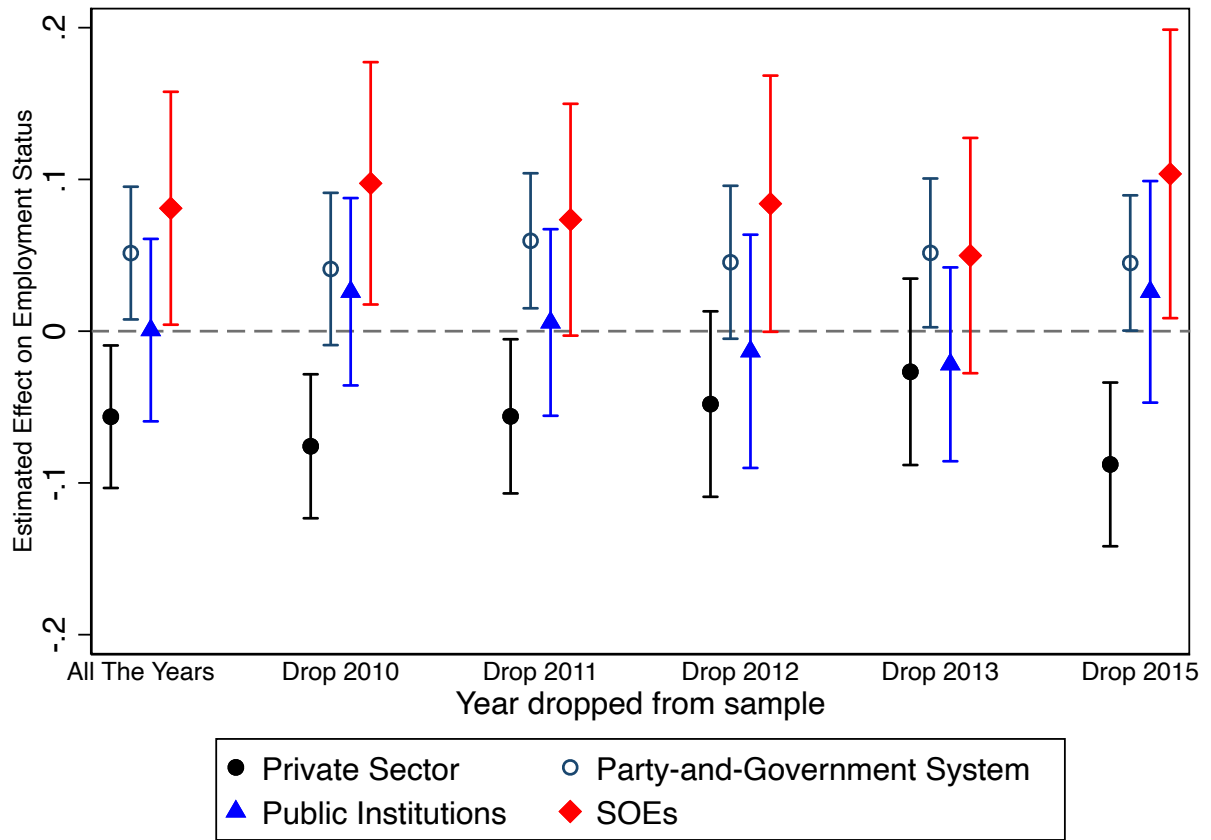
(c) Perception of Increased Compensation & Welfare associated with the VAT Reform, by Social Status



(d) Perception of Increased Compensation & Welfare associated with the VAT Reform, by Education Level

Notes: This figure presents the local polynomial smoothed conditional probabilities (means) of individuals in different social classes or with different levels of educational attainment perceiving increases in (i) *power*, and (ii) *compensation and welfare*, respectively, associated with the SAT agencies after the VAT reform. The sample includes individuals with at least short-course college education level (“大学专科”). In Subfigures (a) and (b), the solid line represents the conditional probabilities of perceiving increased power associated with SAT agencies. In Subfigures (c) and (d), the solid line represents the conditional probabilities of perceiving increased compensation and welfare associated with SAT agencies. The shaded areas indicate the estimated conditional probabilities’ 95% confidence intervals. In Subfigures (a) and (c), the independent variable (x-axis) “Social status” is a continuous, self-reported measure of an individual’s social class and status, quantified on an 11-point scale ranging 0 (the lowest status) to 10 (the highest class). In subfigures (b) and (d), the independent variable (x-axis) is the level of educational attainment - 5 (College with short-cycle courses, 大学专科), 6 (College with normal courses, 大学本科), 7 (Master degrees, 硕士), and 8 (Doctoral degrees, 博士).

Figure A6: The Local Polynomial Smoothed Conditional Mean (Probability): Perceived Power and Benefit Change associated with the VAT Reform, across Different Education Levels and Social Statuses



Notes: This figure presents the sensitivity of CGSS results for the NCSE-eligible group to the composition of survey years. Our preferred specification for these estimates controls for province-level log GDP, individual characteristics (gender, political identity, ethnicity, and residence), year fixed effects, and a full set of interactions of province-by-cohort fixed effects. For each dependent variable - working in the private sector, working in the party-and-government system, working in public institutions, and working in SOEs - we re-estimate our preferred regression specification after dropping one survey year at a time (except 2017, the only post-reform wave), *using the NCSE-eligible sample*. The first set of plots (marked as “All The Years”) are based on all the years of the NCSE-eligible sample. Dots denote point estimates of the coefficient on the interaction between the High-BT-VAT indicator and the post-reform dummy, and vertical bars show their 95% confidence intervals.

Figure A7: Estimated Effects of VAT Reform on Individual Employment Status (including Private Sector, Party-and-Government System, Public Institutions, and State-Owned Enterprises) for the *NCSE-Eligible Group*, CGSS

A3 Appendix Tables

Table A1 presents the summary statistics of the treatment and outcome variables by the two subsamples. The average prefecture-level BT-VAT ratio in our sample is 2.16. Overall, there are more job openings in the SAT sample versus the non-SAT sample. This is because the SAT has the largest body of local branches among all central government agencies. Within the SAT sample, the number of applicants per position is close to 100 in the past 12 years, with about 15 more applicants per position in the high BT-VAT regions than in the low BT-VAT regions on average. The average App-to-Recruit ratio and the interview cutoff score are also higher in the high BT-VAT regions than in the low BT-VAT regions. These patterns are similar in the non-SAT sample.

Table A1: Summary Statistics. NCSE 2010-2021

	All		Low BT-VAT		High BT-VAT	
	Obs.	Mean	Obs.	Mean	Obs.	Mean
Pref. BT-VAT Ratio (2011)	288	2.164	175	1.393	113	3.357
Panel A: SAT Sample						
# of Applicants	75,498	98.56	40,100	93.18	35,398	104.7
App-to-Recruit	75,498	63.12	40,100	61.53	35,398	64.92
Min. Score for Interview	56,892	120.5	29,486	120.1	27,406	120.9
Recruitment Quota	75,881	1.761	40,237	1.702	35,644	1.828
Panel B: Non-SAT Sample						
# of Applicants	61,747	122.9	24,258	92.33	37,489	142.8
App-to-Recruit	61,747	85.12	24,258	72.05	37,489	93.58
Min. Score for Interview	41,794	119.4	16,615	118.7	25,179	119.9
Recruitment Quota	64,504	1.698	24,936	1.486	39,568	1.832

Notes: The BT-VAT Ratio (2011) is measured at the prefecture level (288 prefectures). Observations with missing and extreme values of BT-VAT ratios are excluded. Other variables are measured at *job opening-year* levels. The minimum entry scores for interviews are normalized to a 200-point scale for all the job openings. The SAT-sample includes job openings affiliated with the SAT agencies. The non-SAT-sample includes job openings outside the SAT agencies. The low BT-VAT sample include 175 prefectures with below-mean BT-VAT ratio in 2011. The high BT-VAT sample include 113 prefectures with above-mean BT-VAT ratio in 2011.

Table A2 shows the timeline of the gradual roll-out of the VAT reform. The VAT reform started in 2012 with a pilot only in Shanghai. In the same year, the pilot was expanded to two more central-administered municipalities, Beijing and Tianjin, and other eight provinces. In these pilot regions, the BT was replaced with the VAT partly in transportation and modern service industries. From August 2013 to June 2014, the central government expanded the VAT reform to the rest

of the country, but only covers a few small industries as pilots, such as the transportation and modern service industries and part of radio, film, and television products and telecommunication industries. In May 2016, the VAT reform was fully implemented for the entire country and covered all industries.

Table A2: The Timeline of Experimentation and Completion of VAT reforms in China

Time	Experimented Locations	Experimented Industries
Jan 2012	Shanghai	Part of transportation and modern service industries
Sept 2012	Shanghai, Beijing	Part of transportation and modern service industries
Oct 2012	Shanghai, Beijing, Jiangsu, Anhui,	Part of transportation and modern service industries
Nov 2012	Shanghai, Beijing, Jiangsu, Anhui, Fujian, Guangdong	Part of transportation and modern service industries
Dec 2012	Shanghai, Beijing, Jiangsu, Anhui, Fujian, Guangdong, Tianjin, Zhejiang, Hubei	Part of transportation and modern service industries
Aug 2013	Nationwide	Part of transportation and modern service industries + Radio, film, and television products
Jan 2014	Nationwide	All of transportation and modern service industries + Radio, film, and television products
June 2014	Nationwide	All of transportation and modern service industries + Radio, film, and television products + Telecom industries
May 2016	Nationwide	All of transportation and modern service industries + Radio, film, and television products + Telecom industries + Housing and real estates, finance, construction, consumer service industries

Table A3: Effects of the VAT Reform on NCSE Outcomes with Continuous BT-VAT Ratio Treatment

SAT Sample	# of Applicants	App-to-Recruit Ratio	Min. Score for Interviews
	(1)	(2)	(3)
D.V. Mean	97.07	61.29	120.5
BT-VAT 2011 Ratio $\times$ Post2016	8.961* (5.041)	4.746** (2.013)	0.474** (0.185)
Observations	70,498	70,498	56,086
R^2	0.105	0.094	0.422
Prefecture FEs	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
Prefecture Covariates	Yes	Yes	Yes
Position Covariates	Yes	Yes	Yes
Number of Provinces	31	31	31
Number of Prefectures	288	288	288

Notes: Results are based on the SAT sample. “# of Applicants” is the number of applicants for each position. “App-to-Recruit” is the abbreviation of position-level ratio of the number of applicants to recruitment quota. “Min. Interview Score” is the minimum entry scores for interviews, normalized to a 200-point scale for all the job positions. BT-VAT Ratio 2011 is a prefecture’s 2011 BT-VAT ratio. *Post2016* equals 1 for all years later than 2016 and 0 otherwise. Prefecture fixed effects, year fixed effects, prefecture-level covariates, and position-level covariates are included across all the specifications. Standard errors reported in parentheses are clustered at the prefecture level: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Effects of the VAT Reform on NCSE Outcomes with Alternative Treatment Assignment

SAT Sample	# of Applicants		App-to-Recruit Ratio		Min. Score for Interviews	
	(1)	(2)	(3)	(4)	(5)	(6)
D.V. Mean	98.56	97.07	63.12	61.29	120.5	120.5
I(Ave. BT-VAT > Mean) $\times$ Post2016	17.738* (9.692)	15.813* (9.420)	10.368** (4.816)	9.063** (4.567)	0.994** (0.401)	1.090*** (0.408)
Observations	75,498	70,498	75,498	70,498	56,892	56,086
R^2	0.068	0.104	0.059	0.093	0.357	0.422
Prefecture FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Prefecture Covariates		Yes		Yes		Yes
Position Covariates		Yes		Yes		Yes
Number of Provinces	31	31	31	31	31	31
Number of Prefectures	288	288	288	288	288	288

Notes: Results are based on the SAT sample. “# of Applicants” is the number of applicants for each position. “App-to-Recruit” is the abbreviation of position-level ratio of the number of applicants to recruitment quota. “Min. Interview Score” is the minimum entry scores for interviews, normalized to a 200-point scale for all the job positions. “I(Ave. BT-VAT > Mean)” equals 1 if a prefecture’s average BT-VAT ratio during years from 2007 to 2011 is higher than that period’s cross-prefecture average. *Post2016* equals 1 for all years later than 2016 and 0 otherwise. Prefecture fixed effects, year fixed effects, prefecture-level covariates, and position-level covariates are included across all the specifications. Standard errors reported in parentheses are clustered at the prefecture level: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We examine whether our results are affected by the 2012 pilot reform that introduced the VAT in selected provinces. Although this pilot — implemented in transportation and related sectors — affected only a small fraction of the economy, we allow pilot and non-pilot provinces to have differential outcome changes before and after 2012 in Table A5. Controlling for this leaves our main results unchanged, and the coefficient on the pilot reform control is statistically insignificant.

Table A5: Effects of the VAT Reform on NCSE Outcomes with Controls for Pilot VAT Experimentation

Sample: SAT	# of Applicants		App-to-Recruit Ratio	
	(1)	(2)	(3)	(4)
D.V. Mean	98.56	97.07	63.12	61.29
High BT-VAT × <i>Post2016</i>	18.916** (9.486)	20.085** (9.238)	10.902** (4.817)	11.180** (4.614)
Pilot Province × <i>post2012</i>	11.300 (9.886)	3.252 (8.535)	-2.869 (7.249)	-8.444 (6.184)
Observations	75,498	70,498	75,498	70,498
R^2	0.068	0.104	0.059	0.094
Prefecture FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Pref. Covar.		Yes		Yes
Post. Chara. FEs		Yes		Yes
Number of Provinces	31	31	31	31
Number of Prefectures	288	288	288	288

Notes: Results are based on the SAT sample. “# of Applicants” is the number of applicants for each position. “App-to-Recruit” is the abbreviation of position-level ratio of the number of applicants to recruitment quota. Because the data of the minimum entry score for interviews is only available in and after 2014, this test cannot use it as the dependent variable. “High BT-VAT” equals 1 if a prefecture’s 2011 BT-VAT ratio is higher than the average. *Post2016* equals 1 for all years later than 2016 and 0 otherwise. “Pilot Province” equals 1 if the observation is in the following provinces or central-administered province-level municipalities: Beijing, Shanghai, Jiangsu, Anhui, Fujian, Guangdong, Tianjin, Zhejiang, and Hubei. *Post2012* equals 1 for all years later than 2012 and 0 otherwise. Prefecture fixed effects, year fixed effects, prefecture-level covariates, and position-level covariates are included across all the specifications. Standard errors reported in parentheses are clustered at the prefecture level: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In Table A6, Columns (1), (3), and (5) control for the full set of interactions between province and year dummies (province-specific heterogeneous trends). To show that our treatment variable does *not* just capture the boom in the service sector, we control for the full set of interactions of prefecture-level covariates - GDP per capita, Log population, secondary (manufacturing) sector's GDP share, and tertiary (service) sector's GDP share - in 2011 (one year prior to the pilot) and year dummies, in Columns (2), (4), and (6). Our results are robust to all these specification changes.

Table A6: Effects of the VAT Reform on NCSE Outcomes, Robustness Checks: Controlling for Heterogeneous Trends across Provinces or Trends Correlated with Baseline Covariates

Sample: SAT	# of Applicants		App-to-Recruit Ratio		Min. Score for Interview	
	(1)	(2)	(3)	(4)	(5)	(6)
D.V. Mean	97.03	97.36	61.28	61.32	120.5	120.6
High BT-VAT $\times$ Post2016	11.15** (4.867)	12.27** (5.593)	7.878*** (2.768)	6.682** (3.122)	0.896*** (0.279)	0.857** (0.351)
Observations	70,472	70,445	70,472	70,445	56,086	56,150
R^2	0.153	0.154	0.137	0.140	0.457	0.455
Prefecture FEs	Yes	Yes	Yes	Yes	Yes	Yes
(Province $\times$ Year) FEs	Yes	Yes	Yes	Yes	Yes	Yes
Time-Varying Pref. Covar.	Yes		Yes		Yes	
2011 Pref. Log GDP per capita $\times$ Year Dummies		Yes		Yes		Yes
2011 Pref. Log Population $\times$ Year Dummies		Yes		Yes		Yes
2011 Pref. 2nd Sector GDP Share $\times$ Year Dummies		Yes		Yes		Yes
2011 Pref. Tertiary Sector GDP Share $\times$ Year Dummies		Yes		Yes		Yes
Position Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Number of Provinces	31	31	31	31	31	31
Number of Prefectures	288	285	288	285	288	285

Notes: Results are based on the SAT sample. “# of Applicants” is the number of applicants for each position. “App-to-Recruit” is the abbreviation of position-level ratio of the number of applicants to recruitment quota. “Min. Interview Score” is the minimum entry scores for interviews, normalized to a 200-point scale for all the job positions. “High BT-VAT” equals 1 if a prefecture’s 2011 BT-VAT ratio is higher than the average. *Post2016* equals 1 for all years later than 2016 and 0 otherwise. All the columns control for prefecture fixed effects, position-level covariates, and a full set of interactions between province and year fixed effects. Columns (1), (3), and (5) control for time-varying prefecture-level covariates, including log GDP per capita, log population of residence, and the combined GDP share of the secondary (manufacturing) and tertiary (service) sectors. Columns (2), (4), and (6) add the full set of interactions between the 2011 prefecture-level covariates (including prefecture-level log GDP per capita, log population, secondary sector GDP share, and tertiary GDP share in 2011) and year dummies. Standard errors reported in parentheses are clustered at the prefecture level: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7 shows that our results are robust even when the sample is restricted to years before 2019. Thus, our main results are unlikely to be driven by the 2019 integration of Local Tax Administrations into the SAT. The Chinese central government proposed this integration towards the end of 2018, and it was not expected to take effect until 2019. This robustness check also suggests that our estimation is robust to confounding shocks from the COVID-19 pandemic.

Table A7: Effects of the VAT Reform on NCSE Outcomes, before SAT-Local Tax Administration Integration (2019) and the COVID-19 Pandemic (2020 and after)

Sample: SAT (2019-2021 excluded)	# of Applicants	App-to-Recruit Ratio	Min. Score for Interview
	(1)	(2)	(3)
D.V. Mean	90.34	61.47	119.1
High BT-VAT $\times$ Post2016	18.200** (7.291)	11.144** (5.347)	0.863* (0.451)
Observations	47,547	47,547	32,824
R^2	0.118	0.120	0.393
Prefecture FEs	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
Pref. Covar.	Yes	Yes	Yes
Position Covariates	Yes	Yes	Yes
Number of Provinces	31	31	31
Number of Prefectures	288	288	286

Notes: These results are based on the SAT sample before 2019 (i.e., excluding observations in years 2019, 2020, and 2021). “# of Applicants” is the number of applicants for each position. “App-to-Recruit” is the abbreviation of position-level ratio of the number of applicants to recruitment quota. “Min. Interview Score” is the minimum entry scores for interviews, normalized to a 200-point scale for all the job positions. “High BT-VAT” equals 1 if a prefecture’s 2011 BT-VAT ratio is higher than the average. *Post2016* equals 1 for all years later than 2016 and 0 otherwise. Prefecture fixed effects, year fixed effects, prefecture-level covariates, and position-level covariates are included across all the specifications. Standard errors reported in parentheses are clustered at the prefecture level: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Effects of the VAT Reform on NCSE Outcomes, excluding transition years of reform (2016 and 2017)

Sample: SAT (2016 & 2017 excluded)	# of Applicants	App-to-Recruit Ratio	Min. Score for Interview
	(1)	(2)	(3)
D.V. Mean	99.93	63.96	120.4
High BT-VAT $\times$ Post2016	22.745** (10.573)	10.298** (5.011)	1.428*** (0.427)
Observations	54,463	54,463	40,107
R^2	0.106	0.096	0.466
Prefecture FEs	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
Pref. Covar.	Yes	Yes	Yes
Position Covariates	Yes	Yes	Yes
Number of Provinces	31	31	31
Number of Prefectures	288	288	288

Notes: These results are based on the SAT sample excluding observations in years 2016 and 2017 - the transition years of reform. “# of Applicants” is the number of applicants for each position. “App-to-Recruit” is the abbreviation of position-level ratio of the number of applicants to recruitment quota. “Min. Interview Score” is the minimum entry scores for interviews, normalized to a 200-point scale for all the job positions. “High BT-VAT” equals 1 if a prefecture’s 2011 BT-VAT ratio is higher than the average. *Post2016* equals 1 for all years later than 2016 and 0 otherwise. Prefecture fixed effects, year fixed effects, prefecture-level covariates, and position-level covariates are included across all the specifications. Standard errors reported in parentheses are clustered at the prefecture level: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Effects of the VAT Reform on NCSE Outcomes, Comparing SAT and Other Agencies, within Prefectures by Years

Sample: SAT & Non-SAT	# of Applicants	App-to-Recruit Ratio	Min. Score for Interview
	(1)	(2)	(3)
D.V. Mean	106.9	70.98	120.1
SAT $\times$ High BT-VAT $\times$ Post2016	38.543** (17.044)	18.964** (8.436)	0.831 (0.506)
Observations	128,323	128,323	97,554
R^2	0.119	0.128	0.402
(Prefecture $\times$ Year) FEs	Yes	Yes	Yes
(SAT $\times$ Year) FEs	Yes	Yes	Yes
(SAT $\times$ Prefecture) FEs	Yes	Yes	Yes
Position Covariates	Yes	Yes	Yes
Number of Provinces	31	31	31
Number of Prefectures	288	288	288

Notes: These results are estimated when pooling SAT and non-SAT samples together. “# of Applicants” is the number of applicants for each position. “App-to-Recruit” is the abbreviation of position-level ratio of the number of applicants to recruitment quota. “Min. Interview Score” is the minimum entry scores for interviews, normalized to a 200-point scale for all the job positions. “High BT-VAT” equals 1 if a prefecture’s 2011 BT-VAT ratio is higher than the average. *Post2016* equals 1 for all years later than 2016 and 0 otherwise. “SAT” is the SAT dummy which equals 1 if the position is in an SAT agency and 0 if it is in a non-SAT agency. All specifications include province-by-year fixed effects, interactions between the SAT indicator and year dummies (SAT $\times$ Year fixed effects), interactions between the SAT indicator and prefecture dummies (SAT $\times$ Prefecture fixed effects), as well as position-level covariates. These interactions of fixed effects absorb all the constitutive linear terms (SAT, High BT-VAT, and Post2016) and all the constitutive interaction terms (SAT $\times$ Post2016, SAT $\times$ High BT-VAT, and High BT-VAT $\times$ Post2016). Standard errors reported in parentheses are clustered at the prefecture level: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Social Elites and Individual Perception of Power and Benefits

	Power: Increase			Compensation & Welfare: Increase		
	OLS (1)	OLS (2)	Probit (3)	OLS (4)	OLS (5)	Probit (6)
D.V. Mean	0.627	0.634	0.635	0.705	0.708	0.699
Social status	0.036*** (0.008)	0.044*** (0.010)	0.125*** (0.031)	0.028 (0.017)	0.026 (0.020)	0.078 (0.062)
College or above	0.189** (0.071)	0.155* (0.078)	0.416** (0.206)	0.042 (0.074)	0.031 (0.101)	0.110 (0.308)
Age		-0.002 (0.003)	-0.006 (0.008)		-0.003 (0.004)	-0.008 (0.011)
Female		-0.055 (0.055)	-0.136 (0.151)		0.136** (0.050)	0.423*** (0.152)
CCP member		-0.089 (0.081)	-0.249 (0.227)		0.069 (0.070)	0.208 (0.230)
Married		0.072 (0.055)	0.200 (0.150)		-0.091* (0.048)	-0.283* (0.153)
Minority		-0.002 (0.143)	0.006 (0.384)		-0.216 (0.130)	-0.586* (0.344)
Observations	453	415	411	444	407	392
R-squared	0.058	0.116		0.019	0.110	
Pseudo R-squared			0.0825			0.0759
Province FEs	No	Yes	Yes	No	Yes	Yes
Number of Provinces	28	27	23	28	27	21

Notes: The sample includes individuals with at least short-course college education level (“大专学历”). The dependent variable “Power: Increase” for Columns (1), (2) and (3) is a dummy variable which equals 1 if an individual perceived that the VAT reform increased the power of government positions affiliated with the SAT agencies, and 0 otherwise. The dependent variable “Compensation & Welfare: Increase” for Columns (4), (5) and (6) is a dummy variable which equals 1 if an individual perceived that the VAT reform increased the welfare and compensation associated with government positions affiliated with the SAT agencies, and 0 otherwise. “Social status” is a continuous, self-reported measure of an individual’s social class and status, quantified on an 11-point scale ranging 0 (the lowest status) to 10 (the highest class). “College or above” is a dummy variable which equals 1 if an individual had a bachelor’s degree or above, and 0 if only short-course college education. “Age” is a continuous measure of individuals’ age, which equals 2023 minus the numeric values of their birth years. “CCP member” is a dummy variable, which equals 1 if an individual is a CCP member, and 0 otherwise. “Married” is a dummy variable, which equals 1 if an individual is married, and 0 otherwise. “Minority” is a dummy variable, which equals 1 if an individual is an ethnic minority, and 0 otherwise. Columns (1), (2), (4) and (5) present OLS estimates, while Columns (3) and (6) present Probit estimates. Province fixed effects are added in Columns (2), (3), (5) and (6). Standard errors reported in parentheses are clustered at the province level: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The Provincial Civil Service Exam (PCSE) is responsible for the recruitment of Local Administration of Tax (LAT). The PCSE is held by each province and each province's PCSE information is not as well documented as the NCSE. Moreover, many provinces do not have data for pre-2016 periods. The PCSE dataset we use is from Zhejiang province, which contains all the positions from years 2014, 2015, 2016 and 2018. Table A11 presents the summary statistics of the treatment and outcome variables for the PCSE dataset from Zhejiang Province.

Table A11: Summary Statistics. PCSE 2014, 2015, 2016, 2018

	All		Low BT-VAT		High BT-VAT	
	Obs.	Mean	Obs.	Mean	Obs.	Mean
Pref. BT-VAT Ratio (2011)	11	1.901	9	1.696	2	2.826
Panel A: SAT Sample						
# of Applicants	1,078	67.66	930	60.35	148	113.7
App-to-Recruit	1,078	35.26	930	33.17	148	48.38
Recruitment Quota	1,078	1.764	930	1.726	148	2.007

Notes: The BT-VAT Ratio (2011) is measured at the prefecture level (11 prefectures) in Zhejiang Province. Other variables are measured at *job opening-year* levels. The minimum entry scores for interviews are absent for this PCSE data. The SAT-sample includes job openings affiliated with the SAT agencies. Within Zhejiang Province, the low BT-VAT sample include 9 prefectures with below-mean BT-VAT ratio in 2011, while the high BT-VAT sample include 2 prefectures with above-mean BT-VAT ratio in 2011.

Table A12: Effects of the VAT Reform on NCSE Outcomes: Heterogeneous Analysis (Alternative Lists of Keywords to Identify “Powerful Positions”)

Panel A	# of applicants	App-to-Recruit Ratio	# of applicants	App-to-Recruit Ratio
Sample: SAT				
	Powerful position (Keyword “ <i>zhi fa</i> ” excluded)		Powerful position (Keyword “ <i>shui yuan</i> ” excluded)	
	(1)	(2)	(3)	(4)
D.V. Mean	102	64.64	81.02	57.26
High BT-VAT × Post2016	32.150* (16.887)	21.848** (8.830)	38.362* (19.638)	19.090 (12.851)
Observations	7,013	7,013	3,446	3,446
R^2	0.152	0.139	0.245	0.219
Number of Provinces	29	29	27	27
Number of Prefectures	176	176	145	145
Panel B	# of applicants	App-to-Recruit Ratio	# of applicants	App-to-Recruit Ratio
Sample: SAT				
	Powerful position (Keyword “ <i>ji cha</i> ” excluded)		Powerful position (Keyword “ <i>shen ji</i> ” excluded)	
	(5)	(6)	(7)	(8)
D.V. Mean	104.7	67.55	98.95	64.27
High BT-VAT × Post2016	-2.847 (16.353)	0.167 (10.646)	31.960* (17.090)	22.452** (8.850)
Observations	5,117	5,117	7,625	7,625
R^2	0.162	0.142	0.156	0.140
Number of Provinces	23	23	29	29
Number of Prefectures	142	142	185	185
Controls:				
Prefecture FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Pref. Covar.	Yes	Yes	Yes	Yes
Post. Chara. FEs	Yes	Yes	Yes	Yes

Notes: Results are based on the SAT sample of “powerful positions”, using different definitions based on alternative lists of keywords. “# of Applicants” is the number of applicants for each position. “App-to-Recruit” is the abbreviation of position-level ratio of the number of applicants to recruitment quota. “High BT-VAT” equals 1 if a prefecture’s 2011 BT-VAT ratio is higher than the average. *Post2016* equals 1 for all years later than 2016 and 0 otherwise. We exclude one specific keyword each time from the list of keywords which define the “powerful positions”: we exclude “*zhi fa*” (“tax law enforcement”, 执法) in Columns (1) and (2), exclude “*shui yuan*” (“tax base”, 税源) in Columns (3) and (4), exclude “*ji cha*” (“tax inspections”, 稽查) in Columns (5) and (6), and exclude “*shen ji*” (“tax audit”, 审计) in Columns (7) and (8). Prefecture fixed effects, year fixed effects, prefecture-level covariates, and position-level covariates are included across all the specifications. Standard errors reported in parentheses are clustered at the prefecture level: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A Potential Mechanism: Graft Opportunities In Table A13, we examine how the effect of the VAT reform varies with regional differences in historical corruption levels. We construct a prefecture-level measure of corruption cases based on the anti-corruption campaign database from Y. Wang and Dickson (2022). The database, compiled by Tencent, includes information on convicted government officials such as their names, positions, localities, ranks, and the grounds for investigation. In 2012, Chinese President Xi Jinping launched an anti-corruption campaign that is still ongoing. Cases investigated in a prefecture after 2012 may reflect the intensity of the crack-down and potentially political purges in that region. In contrast, the number of corruption cases in a prefecture before 2012 reflects corruption opportunities in that region, since investigations before this period were less related to political purges or factional politics. Thus, using pre-2012 corruption cases is more likely to capture citizens' perceived opportunities for corruption. We use the accumulated corruption-investigation cases (2004 - 2012) aggregated at the prefecture level.

Table A13 shows that, following the reform, SAT positions in regions with higher perceived opportunities for corruption, as indicated by the number of past corruption cases, attract significantly more applicants and exhibit higher applicant-to-recruit ratios. Furthermore, SAT positions in these more corruption-prone areas also show higher passing scores, indicating that these positions are being filled by candidates with higher measured abilities. These results provide further evidence that state building attracts talent by increasing the graft opportunities of government jobs.

In Table A14, we interact the VAT reform treatment with a prefecture's pre-reform share of state-owned enterprise (SOE) output in total enterprise output. This share is calculated from the 2008 Chinese Industrial Enterprise Survey (CIES). The CIES has covered all the enterprises above the designated size —defined as industrial firms with annual main business revenue of at least RMB 5 million. By dividing the total output of recorded SOEs by the total output of all the recorded enterprises in each prefecture, we obtain the SOE's output share at the prefecture level. We use this variable to measure the degree of SOE presence in the economy. We find that the number of applications and competition for the SAT job positions in prefectures with higher SOE presence experienced a greater increase in response to the VAT reform.

Table A13: Effects of the VAT Reform on NCSE Outcomes: Heterogeneity based on Corruption Cases

Sample: SAT	# of Applicants	App-to-Recruit Ratio	Min. Score for Interview
	(1)	(2)	(3)
D.V. Mean	97.14	61.36	120.5
High BT-VAT $\times$ Post2016 $\times$ Corruption	23.817*** (5.522)	8.184** (3.247)	0.613** (0.238)
High BT-VAT $\times$ Post2016	3.026 (7.787)	5.552 (4.275)	0.879** (0.427)
Corruption $\times$ Post2016	-6.252** (3.122)	-2.705 (1.970)	-0.263 (0.201)
N of Observations	70,334	70,334	55,955
R^2	0.108	0.094	0.422
Prefecture FEs	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
Prefecture Covariates	Yes	Yes	Yes
Position Characteristics FEs	Yes	Yes	Yes
Number of Provinces	31	31	31
Number of Prefectures	287	287	287

Notes: These results are based on the SAT sample. “# of Applicants” is the number of applicants for each position. “App-to-Recruit Ratio” is the position-level ratio of the number of applicants to recruitment quota. “Min. Score for Interview” is the minimum entry score for interviews, normalized to a 200-point scale for all job positions. “High BT-VAT” equals 1 if a prefecture’s 2011 BT-VAT ratio is higher than the average. *Post2016* equals 1 for all years later than 2016 and 0 otherwise. “Corruption” is the cumulative corruption cases from 2004 to 2012) aggregated at the prefecture level. The data is from [Y. Wang and Dickson \(2022\)](#). Prefecture fixed effects, year fixed effects, prefecture-level covariates, and position characteristics fixed effects are included across all specifications. The constitutive term *Corruption* is absorbed by prefecture fixed effects and *Post2016* absorbed by year fixed effects. Standard errors reported in parentheses are clustered at the prefecture level: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A14: Effects of the VAT Reform on NCSE Outcomes: Heterogeneity based on Local SOE Output Shares

Sample: SAT	# of Applicants	App-to-Recruit Ratio	Min. Score for Interview
	(1)	(2)	(3)
D.V. Mean	97.08	61.44	120.7
High BT-VAT $\times$ Post2016 $\times$ Prefecture SOE Share	138.496** (65.141)	58.726* (31.234)	2.428 (2.663)
Prefecture SOE Share $\times$ Post2016	-45.531** (20.378)	-24.733** (11.397)	-1.326 (1.145)
High BT-VAT $\times$ Post2016	-4.398 (13.179)	0.873 (6.660)	0.874 (0.687)
Observations	69,437	69,437	55,305
R^2	0.104	0.094	0.416
Prefecture FEs	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
Pref. Covar.	Yes	Yes	Yes
Post. Chara. FEs	Yes	Yes	Yes
Number of Provinces	29	29	29
Number of Prefectures	282	282	282

Notes: These results are based on the SAT sample. “# of Applicants” is the number of applicants for each position. “App-to-Recruit” is the abbreviation of position-level ratio of the number of applicants to recruitment quota. “Min. Interview Score” is the minimum entry scores for interviews, normalized to a 200-point scale for all the job positions. “High BT-VAT” equals 1 if a prefecture’s 2011 BT-VAT ratio is higher than the average. *Post2016* equals 1 for all years later than 2016 and 0 otherwise. “Prefecture SOE Share” is the output share of state-owned enterprises at the prefecture level, calculated from the 2008 China Industrial Enterprise Survey (CIES) data. Prefecture fixed effects, year fixed effects, prefecture-level covariates, and position-level covariates are included across all the specifications. The constitutive terms *Prefecture SOE Share*, *High BT-VAT*, and *Prefecture SOE Share $\times$ High BT-VAT* are absorbed by prefecture fixed effects, and *Post2016* are absorbed by year fixed effects, respectively. Standard errors reported in parentheses are clustered at the prefecture level: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In Table A15, we control for province-level log GDP, individual characteristics, and the full set of interactions of province-by-cohort fixed effects. Thus, the variations that we use are from comparisons between years before and after the reform for the same birth-year cohort within the same province, and comparisons between provinces with high and low BT-VAT ratios in the same year.

Table A15: The Effect of the VAT Reform on Public Sector Employment, CGSS, by (i) Party-and-Government System, (ii) Public Institutions, and (ii) State-Owned Enterprises

Sample: CGSS	NCSE-eligible			Placebo: NCSE-non-eligible		
	Work in Party-and Government (1)	Work in Public Institutions (2)	Work in SOEs (3)	Work in Party-and Government (4)	Work in Public Institutions (5)	Work in SOEs (6)
D.V. Mean (For estimated sample)	0.0850	0.292	0.245	0.0510	0.140	0.152
D.V. Mean (For low BT-VAT provinces in pre-reform years in estimated sample)	0.119	0.366	0.264	0.0473	0.139	0.138
High BT-VAT × Post2016	0.051** (0.022)	0.001 (0.031)	0.081** (0.039)	0.008 (0.014)	0.017 (0.021)	0.028 (0.018)
Observations	2,140	2,824	2,608	13,749	15,185	15,395
R ²	0.338	0.279	0.196	0.230	0.212	0.211
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
LnGDP	Yes	Yes	Yes	Yes	Yes	Yes
Ind. Charac.	Yes	Yes	Yes	Yes	Yes	Yes
Province × Cohort FEs	Yes	Yes	Yes	Yes	Yes	Yes
Number of Provinces	27	27	26	28	28	28

Note: The data is from the 2010, 2011, 2012, 2013, 2015, and 2017 waves of the CGSS (Chinese General Social Surveys). Columns (1), (2), and (3) of this table use the subsample of respondents who meet two eligibility conditions: (i) the education level is college or above; (ii) age is between 18 and 35 (18 and 35 included). Columns (4), (5), and (6) use the subsample of respondents who are either education-non-eligible or age-non-eligible. “Cohort” is defined based on respondents’ birth years. “Ind. Charac.” controls individual characteristics including gender (Male = 1), political identity (Chinese Communist Party member = 1), ethnicity (Han = 1), and residence (Urban = 1). The dependent variable of Columns (1) and (4) equals 1 if respondents self-reported that they work in the “party-and-government” system (“党政机关”), and 0 otherwise; the dependent variable of Columns (2) and (5) equals 1 if respondents self-reported that they work in public institutions (“事业单位”), and 0 otherwise; the dependent variable of Columns (3) and (6) equals 1 if respondents self-reported that they work in SOEs, including state-owned enterprises (“国有或国有控股”) or collectively-owned enterprises (“集体所有或集体控股”), and 0 otherwise. All the specifications control for province-level log GDP, individual characteristics, year fixed effects, and the full set of interactions between province and cohort dummies. In every column, we report dependent variable’s mean values (i) for the estimated sample, and (ii) for the control group (prefectures with low BT-VAT ratios) in pre-reform years (2010-2015) in the estimated sample, respectively. Standard errors clustered at the province level are presented in parentheses: *** p<0.01, ** p<0.05, * p<0.1.

A4 Survey Questionnaire

The survey questions regarding the National Civil Service Exam were included in two other online surveys conducted by one of the authors of this paper. The samples for the two surveys were collected by a US-based survey company from July to August, and from August to November 2023, respectively.

We have given careful consideration to ethics. These surveys were approved by the Institutional Review Board (IRB) of the author's home institution. We also adhere to EGAP principles regarding research transparency and the protection of research team staff. Furthermore, we implemented two measures to minimize potential risks to respondents and research team staff. First, we collected only non-identifiable information, which is securely stored on servers outside China. We informed respondents about the protection of privacy and data security at the beginning of the survey and allowed them to exit the survey at any time. Second, the survey questions for this study ask respondents about whether they have taken civil service exams and their understanding of the Value Added Tax Reform, which are not sensitive at all. Together, these measures minimize any potential risks to the survey participants and research staff.

In total, we obtained a sample of 4,585 respondents. These respondents are all adults and come from most provinces in China, except Qinghai, Ningxia, Xinjiang, and Tibet, due to data availability issues in those regions. While this sample is not nationally representative, we employed a quota sampling strategy to target age, gender, education, and geographic distributions as closely as possible.

Below we present survey questions that we use for our analysis in Table 3 and Table A10:

- **(Question: sedu)** Your highest educational attainment is: (您的最高学历[最高已完成的学历]是:)
 - (1) Primary school or below (小学或小学以下)
 - (2) Junior high school (初中)
 - (3) Vocational secondary or high school, or equivalents (职业高中或同等技术学校)
 - (4) Regular senior high school (普通高中)
 - (5) College with short-cycle courses (大学专科)

- (6) College with normal courses (大学本科)
- (7) Master degrees (硕士)
- (8) Doctoral degrees (博士)
- **(Question: sbirthy)** What is your birth year? Please fill in a 4-digit integer. (您的出生年份是[请填写4位整数]?) _____
- **(Question: sprov)** Which province and city are you permanently living? (您目前长期居住的省市是?) _____
 - **(Variable name: sprov_1)** Province (省)
 - **(Variable name: sprov_2)** City (市)
- **(Question: dparty)** What is your political identity? (您的政治面貌是”)
 - (1) CCP member (共产党员)
 - (2) CCP Youth League member (共青团员)
 - (3) Democratic parties’ member (民主党派)
 - (4) Mass (群众)
- **(Question: dclass)** Which social class do you think you are in? 0 means the lowest class, and 10 means the highest class. (您认为您的家庭在中国社会中处于哪个阶级层次? 0表示最底层, 10表示最顶层”)
 - (0)
 - (1)
 - (2)
 - (3)
 - (4)
 - (5)
 - (6)

- (7)
- (8)
- (9)
- (10)
- **(Question: dmarri)** What is your current marital status? (您目前的婚姻状况是”)
 - (1) Unmarried (未婚)
 - (2) In marriage (已婚)
 - (3) Divorced (离异)
 - (4) Widowed (丧偶)
- **(Question: dethn)** What is your ethnic identity? (您的民族是)
 - (1) Han (汉族)
 - (2) Minority (少数民族)
- **(Question: djob)** Do you have a full-time job, currently or before retirement? If you are a student, please choose “No.” (您目前(或退休前)是否有全职的工作?若您是学生, 选择“否”)
 - (1) Yes (是)
 - (2) No (否)
- **(Question: ncse1)** Have you applied for, or are you planning to apply for, the National Civil Service Examination (the NCSE)? (请问您是否曾经参加过, 或正打算参加国家公务员考试?)
 - (1) Yes (是)
 - (2) No (否)
- **(Question: ncse2)** [If “Yes” to **Question ncse1**] Have you applied for, or are you planning to apply for, any government job openings affiliated with the State Administration of Tax (SAT)? (请问, 对于公务员考试, 您是否曾经报考过或计划报考税务系统的职位?)

- (1) Yes (是)
- (2) No (否)
- **(Question: ncse3)** [If “Yes” to **Question ncse2**] Do you know about the Value-Added Tax (VAT) reform? (请问您知道“营改增”这一政策吗?)
 - (1) Yes (是)
 - (2) No (否)
- **(Question: ncse4)** [If “Yes” to **Question ncse3**] Do you think that the VAT reform increased or decreased the power of the SAT agencies? (请问您觉得“营改增”这一政策, 是增加了还是减少了国税部门的权力?)
 - (1) Increase (增加)
 - (2) Decrease (减少)
 - (3) No Change (保持不变)
- **(Question: ncse5)** [If “Yes” to **Question ncse3**] How do you think the VAT reform would affect the welfare and compensation for the bureaucrats affiliated to the SAT agencies? (请问您觉得“营改增”这一政策, 如何影响国税系统工作人员的待遇和福利?)
 - (1) Increase (增加)
 - (2) Decrease (减少)
 - (3) No Change (保持不变)

The NCSE requires that applicants should have at least college education levels (including college with short-cycle courses). Hence, to estimate the results in Table 3, we limit our sample to the following subgroup, whose educational attainment level makes themselves eligible for the NCSE — those respondents who chose one of the last four options in **Question sedu** (with at least a college education level, 大学专科).

The responses to Questions **ncse1**, **ncse2**, **ncse3**, **ncse4**, and **ncse5** are used to estimate the results in Column (1), (2), (3), (4) and (5) of Table 3, respectively.

To estimate the results in Table A10, we further limit our sample to subgroup (A) only, the group who had *at least* short-course college education level (大学专科) when participating in the survey. We calculate respondents' age by subtracting the numeric values of their birth years from 2023.

A5 SAT Agencies: Powerful and Less powerful Positions

In this section, we detail our methodology for identifying relatively powerful positions from the SAT within the NCSE dataset. We elicit information on the nature of work for different job openings from the texts that the following four identification variables contain: the names of job posts (“职位名称”), the names of affiliated agencies or departments (“部门名称”), the names of the agency branches (“用人司局”), and the descriptions of job posts (“职位简介”).

We conduct expert interviews with employees of the State Administration of Taxation (SAT) in China, who possess at least eight years of working experience in tax administration. These interviews are separate field work activities from the survey we introduce in Appendix A4. These interviews did not need IRB review, because they did not involve human subjects, determined by the home institution of one of the authors with the IRB protocol number 7060X. The questions appearing in the interviews are only about the functioning of the institutions in which the interviewees work. No question in the interview is about the interviewees themselves.

During the interviews, we provided them with a comprehensive list of keywords extracted from the texts contained by the above four identification variables in our dataset. The list presented to the experts is shown below.

- Tax Sources & Collection: Tax Base Management, Tax Collection, Tax Collection and Management and Non-tax Revenue Department.
- Tax Services & Administration: Tax Services, Tax Handling Services, Tax Law Enforcement, Tax Business/Operations, Tax Comprehensive Business/Operations, Tax Comprehensive Management Department, Tax Comprehensive Department.
- Tax Inspections & Cases: Tax Inspections, Tax Audit, Case Review/Trial, Case Execution/Enforcement, Tax Management.

- International & Special Taxes: International Tax Affairs, Goods and Services Tax, Horse Racing Taxation, Vehicle Purchase Tax, Vehicle Acquisition Tax, Import and Export Tax, Income Tax

After reviewing the list, these SAT-employee interviewees identified a set of keywords that, based on their experience, signify “powerful” positions. Specifically, the keywords used to identify “powerful” positions include: “**tax base** (税源),” “**tax inspections** (稽查),” “**tax audit** (审计),” and “**tax law enforcement** (执法).”

All the other positions whose identification variables do not include the above four keywords in their texts are categorized as “less powerful.” Furthermore, we regard positions with designated locations in financially challenged areas, remote locations, or poor areas as “less powerful” as well. The geographic locations of such job openings indicate that it is less likely for their roles to be empowered by the VAT reform than those “powerful” positions. To identify these positions with special designated geographic locations, we use information extracted from the texts contained by the above four identification variables that include the following keywords: “financially challenged (财政困难),” “difficult (困难),” “poor county (贫困县),” “poor areas (贫困地区),” “hardship (艰苦),” and “geographically peripheral areas (偏远).”